Characterisation and Quantification of Data Centre Flexibility for Power System Support

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***Abstract*—Data centres can support power systems by coordinating workload scheduling, electrical storage and cooling, but their available flexibility depends on the operating state and required service duration. This paper develops an integrated model of IT workload shifting, uninterruptible power supply (UPS) batteries, cooling and thermal energy storage. Linked daily mixed-integer linear optimisations establish a cost-minimising operating baseline over 2025. A duration-aware assessment then identifies certified feasible power deviations for specified start times while enforcing workload, storage, thermal and post-event recovery constraints. For a hypothetical data centre with 1 MW of rated IT capacity, coordinated operation reduces annual electricity costs by GBP 29,626.91 (5.38%). Resource sensitivities show the largest change in savings when the flexible workload share is varied, followed by UPS and thermal storage energy capacity. On the selected representative day, a 100 kW load reduction is feasible for six hours from noon and one hour from 06:00. Morning reductions rely predominantly on UPS discharge, whereas afternoon reductions rely mainly on workload deferral and associated cooling reductions. Load increases combine workload execution, cooling and storage actions. The results quantify feasible magnitude–duration combinations under the stated model and recovery assumptions; assessment caps and unresolved trials do not establish maximum durations. Lower electricity costs do not imply lower annual electricity consumption or peak import. The framework provides candidate flexibility offers for subsequent assessment against network constraints and service-specific requirements.**

***Keywords*—Data centre, power system flexibility, demand side flexibility, load shifting, thermal inertia**

1. Introduction

GLOBAL digitalisation, propelled by artificial intelligence, is driving the rapid expansion of data centres, whose electricity demand is both substantial and spatially concentrated. This growth coincides with two other major shifts in the energy landscape: the increasing share of variable renewable generation needed for decarbonisation, and the widespread adoption of new demand side technologies such as electric vehicles and heat pumps [1], [2]. The combined effect places significant strain on modern power systems by intensifying peak loads, steepening net load ramps, and aggravating local network constraints. Consequently, there is an escalating need for power system flexibility to maintain the crucial balance between electricity supply and demand.

According to the IEA, global data centre electricity consumption was around 415 TWh in 2024, with projections suggesting this will reach around 945 TWh by 2030, raising their share of global electricity demand from approximately 1.5% to 3% [3]. This expansion is not uniform, with over 80% of the increase expected to originate from the United States and China [3]. Moreover, geographical clustering intensifies the impact on local grids. For instance, data centre consumption already accounts for over 20% of national demand in Ireland and 25% in Virginia, USA [3]. In Great Britain, annual electricity consumption by data centres is forecast by the National Energy System Operator (NESO) in Future Energy Scenarios (FES) 2025 to rise from 7.6 TWh in 2024 to between 30 and 71 TWh by 2050 [4].

To manage the volatility introduced by these trends, power systems require a substantial increase in flexibility services. Traditionally provided by dispatchable fossil fuel generators, this solution is becoming obsolete under decarbonisation mandates. The JRC [5] projects that the European Union's flexibility requirement will rise to 24% of total electricity demand in TWh by 2030 and to 30% by 2050. Similarly, Great Britain's Clean Flexibility Roadmap targets an expansion of flexibility capacity from 25.2 GW to between 54 and 66 GW by 2030 [6]. Demand side flexibility, where consumers actively modulate their electricity use to support the grid, presents a key solution.

While their growth contributes to the challenge, data centres are equipped to provide flexibility. Their shiftable IT workloads, uninterruptible power supply (UPS) systems with battery storage, and inherent thermal inertia as well as cold storage can be leveraged to modulate power consumption without disrupting core services. Each of these internal assets possesses a distinct flexibility profile in terms of magnitude, duration, ramp rate, and recovery time [7]. For example, UPS batteries can offer a nearly instantaneous response, whereas thermal systems are characterised by lower ramp rates, providing a more gradual adjustment [7], [8]. Data centres therefore present a dichotomy: they are both a growing strain on the power system and a potentially crucial asset for grid flexibility.

In the context of power systems, flexibility is defined as the capability of maintaining the balance between generation and demand across all time horizons by adjusting either energy production or consumption in response to sudden variations [1], [2], [4], [9]. Upward flexibility is required when demand exceeds generation and involves a data centre reducing its power consumption from the grid. Conversely, downward

Table I: Nomenclature

Symbol	Definition	Value/Bounds	Unit
IT Workload Modelling (Section 3.1)			
t, s	Time slot an IT job originates in (t) and time slot the job is executed in (s).	Absolute slot indices	-
T	Committed time slots of local day d; T_d below.	92, 96 or 100 slots	-
T^{ext}	Local-day core plus twelve look-ahead slots; T_d^{ext} below.	$\lvert T_d\rvert + 12$ slots	-
k	Index for tranches of a flexible IT job.	[1,2,3,4]	-
K	Set of tranche indices, with each tranche representing a distinct deferral class.	$\{1, 2, 3, 4\}$	-
$W_{t,k}$	Feasible execution window of time slots for tranche k of a job originating at time t.	Varies	-
Δt	Duration of a single time slot.	0.25	hours
D_k	Maximum delay from arrival to the start of the final eligible execution interval for tranche k. Each interval lasts 15 minutes.	$\{2, 4, 8, 12\}$	slots
$P_{\text{idle}}^{\text{IT}}$	Power consumption of IT equipment at idle CPU utilisation.	166.7	kW
$P_{\text{max}}^{\text{IT}}$	Power consumption of IT equipment at maximum CPU utilisation.	1000	kW
u_{max}	Maximum available CPU capacity in any single time slot.	1	-
u_t^{inflex}	CPU utilisation required for the inflexible workload in time slot t.	[0–1]	-
$u_t^{\text{flex,base}}$	Base CPU utilisation required to process the flexible job originating at slot t without any deferral.	[0–1]	-
R_t	Total computational demand (in CPU-hours) for the flexible job originating in time slot t.	Varies	CPU-hours
$\alpha_{t,k}$	Fraction of the total computational demand R_t that is assigned to tranche k.	[0–1]	-
$u(t)$	CPU utilisation at time t, as a fraction of total capacity.	[0–1]	-
$u(t, k, s)$	Decision variable for the CPU utilisation of tranche k from a job originating at t, scheduled for execution in slot s.	≥ 0	-
$P^{\text{IT}}(t)$	Total power consumption of IT equipment at time t.	[0-1000]	kW
$E_{\text{base}}^{\text{IT}}(t)$	Base IT energy consumption at time t.	Varies	kWh
$E_{\text{opt}}^{\text{IT}}(t)$	Optimised IT energy consumption at time t after optimisation.	Varies	kWh
$P_{\text{base}}^{\text{IT}}(t)$	Base IT power consumption at time t.	[0-1000]	kW
$P_{\text{opt}}^{\text{IT}}(t)$	Optimised IT power consumption at time t after optimisation.	[0-1000]	kW
UPS ESS Modelling (Section 3.2)			
$E_{\text{base}}^{\text{UPS}}$	Base rated energy capacity of the UPS battery system.	600	kWh
SoC_{min}	Minimum state of charge, expressed as a fraction of $E_{\text{UPS}}^{\text{base}}$.	0.5	-
SoC_{max}	Maximum state of charge, expressed as a fraction of $E_{\text{UPS}}^{\text{base}}$.	1.0	-
$P_{\text{ch,min}}^{\text{UPS}}$	Minimum allowable charging power of the UPS.	0	kW
$P_{\text{ch,max}}^{\text{UPS}}$	Maximum allowable charging power of the UPS.	270	kW
$P_{\text{disch,min}}^{\text{UPS}}$	Minimum allowable discharging power of the UPS.	0	kW
$P_{\text{disch,max}}^{\text{UPS}}$	Nominal UPS discharge rating; useful discharge also cannot exceed IT demand (at most 1000 kW).	2700	kW
$\eta_{\text{ch}}^{\text{UPS}}$	Charging efficiency of the UPS battery system.	0.82	-
$\eta_{\text{disch}}^{\text{UPS}}$	Discharging efficiency of the UPS battery system.	0.92	-
$E^{\text{UPS}}(t)$	Energy stored in the UPS battery system at the opening boundary of time slot t.	$[300, 600]$	kWh
$P_{\text{ch}}^{\text{UPS}}(t)$	Charging power drawn by the UPS from the grid during time slot t.	≥ 0	kW
$P_{\text{disch}}^{\text{UPS}}(t)$	Discharging power supplied by the UPS during time slot t.	≥ 0	kW
$P_{\text{net}}^{\text{UPS}}(t)$	Net power exchange of the UPS at time t, defined as $P_{\text{ch}}^{\text{UPS}}(t) - P_{\text{disch}}^{\text{UPS}}(t)$.	Varies	kW
$z_{\text{ch}}^{\text{UPS}}(t)$	Binary variable indicating charging status in time slot t.	$\{0, 1\}$	-
$z_{\text{disch}}^{\text{UPS}}(t)$	Binary variable indicating discharging status in time slot t.	$\{0, 1\}$	-
Cooling System Modelling (Section 3.3)			
$\dot{m}_a$	Constant mass flow rate of air circulated by the cooling unit.	100	kg/s
c_{pa}	Specific heat capacity of air at constant pressure.	1.005	kJ/(kg·K)
ρ_a	Density of air.	1.16	kg/m^3
C_{IT}	Total heat capacity of the IT components.	1.788×10^4	kJ/K
C_R	Total heat capacity of the server racks and the air within them.	1.802×10^4	kJ/K
C_{CA}	Total heat capacity of the air in the cold aisle.	2.33×10^3	kJ/K
C_{HA}	Total heat capacity of the air in the hot aisle.	1.17×10^3	kJ/K

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Table I – continued from previous page

Symbol	Definition	Value/Bounds	Unit
G_{cv}	Convective heat conductance coefficient between IT components and rack air.	109	kW/K
G_{cd}	Total heat transfer coefficient from the cold aisle to the outside environment.	4.484	kW/K
κ	Correction factor for the effectiveness of the air mass flow through racks.	0.766	-
COP_{chiller}	coefficient of performance of the chiller.	5	-
$P_{\max}^{\text{Chiller}}$	Maximum electrical power consumption capacity of the chiller.	400	kW
$E_{\max}^{\text{TES}}$	Maximum cooling energy storage capacity of the TES tank.	1000	kWh
$Q_{\max}^{\text{TES-CRAC}}$	Maximum thermal discharging rate of the TES tank.	300	kW
$Q_{\max}^{\text{Chil-TES}}$	Maximum thermal charging rate of the TES tank.	300	kW
$\eta_{\text{ch}}^{\text{TES}}$	Charging efficiency of the TES tank.	0.9	-
$\eta_{\text{dis}}^{\text{TES}}$	Discharging efficiency of the TES tank.	0.9	-
T_{out}	Outside ambient air temperature.	22	°C
$Q^{\text{IT}}(t)$	Heat generated by ITE, assumed equal to $P^{\text{IT}}(t)$.	[0-1000]	kW
$Q_{\text{cool}}(t)$	Total cooling power provided to the data centre from the CRAC and TES.	≥ 0	kW
$Q^{\text{Chil-CRAC}}(t)$	Cooling power provided directly from the chiller to the CRAC system.	Varies	kW
$Q^{\text{Chil-TES}}(t)$	Cooling power from the chiller used to charge the thermal energy storage (TES) tank.	$[0, 300]$	kW
$Q^{\text{TES-CRAC}}(t)$	Cooling power discharged by the TES tank to the CRAC system.	$[0, 300]$	kW
$Q_{CA}(t)$	Rate of change of thermal energy in the air mass of the cold aisle.	Varies	kW
$Q_{HA}(t)$	Rate of change of thermal energy in the air mass of the hot aisle.	Varies	kW
$Q_R(t)$	Rate of change of thermal energy in the mass of the server racks and the air within them.	Varies	kW
$Q_{ITm}(t)$	Rate of change of thermal energy in the thermal mass of the IT equipment itself.	Varies	kW
$Q_{\text{out}}(t)$	Thermal energy transferred into the data centre from the outside ambient environment.	Varies	kW
$P^{\text{Chil-CRAC}}(t)$	Electrical power consumed by the chiller for cooling sent directly to the CRAC.	≥ 0	kW
$P^{\text{Chil-TES}}(t)$	Electrical power consumed by the chiller to charge the TES tank.	≥ 0	kW
$E^{\text{TES}}(t)$	Cooling energy stored in the TES tank at time t.	$[0, 1000]$	kWh
$T_{Ain}(t)$	Air temperature at the inlet of the cold aisle.	$[18, 30]$	°C
$T_{CA}(t)$	Air temperature in the cold aisle.	$[18, 23]$ (all scenarios)	°C
$T_{HA}(t)$	Air temperature in the hot aisle.	$[18, 40]$	°C
$T_R(t)$	Air temperature within the server racks.	$[18, 40]$	°C
$T_{IT}(t)$	Temperature of the IT components.	$[18, 60]$	°C
$z^{\text{Chil-TES}}(t)$	Binary variable indicating TES charging status (1 if charging, 0 otherwise).	$\{0, 1\}$	-
$z^{\text{TES-CRAC}}(t)$	Binary variable indicating TES discharging status (1 if discharging, 0 otherwise).	$\{0, 1\}$	-
Case Studies (Section 4)			
$\pi(t)$	Day-ahead spot electricity price at time t.	Varies	GBP/MWh
P_{tol}	Power tolerance for flexibility provision.	0.1	kW
$P_{\text{IT}}^{\text{Grid}}(t)$	Power drawn from grid for IT load.	≥ 0	kW
$P_{\text{OD}}^{\text{Grid}}$	Constant auxiliary load, assumed to be 7% of a reference mean IT and cooling load of 758.5 kW.	53.095	kW
$P_{\text{base}}^{\text{Grid}}(t)$	Baseline power drawn from grid at time t.	Varies	kW
$P^{\text{Grid}}(t)$	Total power drawn from grid at time t.	Varies	kW
ΔP	Flexibility magnitude (power deviation from baseline).	Varies	kW
τ	Certified feasible duration of flexibility provision, capped by twelve hours and local-day end.	[0–12]	hours
t_0	Start time of flexibility provision, sampled hourly.	[00:00–23:00]	hours

flexibility is needed when generation surpasses demand and is provided by a data centre increasing its electricity consumption to absorb the surplus energy [1], [10]. In this study, a data centre's flexibility is quantified by its capacity for power consumption deviation relative to a baseline and the certified feasible duration for which this deviation can be sustained.

This paper addresses these challenges through two linked contributions. First, we develop a least cost optimisation model for scheduling IT workload, UPS dispatch, and cooling system operation, considering day-ahead electricity prices. Second, and most importantly for power system operation, we introduce a duration-aware flexibility assessment that answers the question: for a requested start time and power deviation, for how long can the data centre sustain that deviation while respecting IT service, storage, thermal and recovery constraints? This turns flexibility from a single capacity estimate into an operationally useful magnitude–duration envelope. Our analysis reveals the asymmetric nature of upward and downward flexibility, detailing the distinct roles of each internal asset. By providing a method to develop an in-depth understanding of data centres' flexibility potential, this work sheds light on how data centres can be more efficiently integrated into our power system.

The main contributions are therefore: (1) a tractable, integrated data centre model that preserves the flexibility-relevant couplings between IT workload execution, UPS operation, cooling power and TES state; (2) a duration-aware assessment that evaluates the feasible flexibility duration across a grid of start times and requested power deviations, including post-event recovery; and (3) a full-year evaluation of operational electricity cost savings, with a component-level cost decomposition and resource sensitivities, including the incremental value of UPS capacity.

The annual evaluation uses linked daily optimisation to avoid basing the economic conclusions on a single price day. This rolling procedure supports the valuation of flexibility rather than constituting a new optimisation algorithm.

The remainder of this paper is structured as follows. Section 2 reviews the existing literature on data centre flexibility. Section 3 details the mathematical modelling of the key data centre components. Section 4 outlines the case studies used to test the model. Section 5 presents and discusses the results, and Section 6 concludes the paper.

1.1 Data Centre Architecture

Data centres are integrated facilities housing IT equipment, such as servers and network devices, which are supported by critical power and cooling infrastructure. As shown in Figure 1, electrical continuity is maintained by uninterruptible power supplies (UPS) with associated battery systems, while cooling systems, including chillers, thermal energy storage (TES) tanks and computer room air conditioning (CRAC) units, manage the substantial heat generated by the servers. Thermal efficiency is often enhanced through optimised airflow strategies like the hot aisle/cold-aisle arrangement.

Each of these core subsystems presents an opportunity for providing demand side flexibility to the power grid. The strategic scheduling of IT workloads, the dispatch of UPS battery storage, and the leveraging of the TES tank and the facility's inherent thermal inertia allow a data centre to modulate its power consumption. The data centre assets are highly interdependent, requiring an integrated approach that considers all of them simultaneously. This is because a change in one subsystem changes the feasible operating range of the others: deferring IT workload reduces both server power and subsequent heat generation, cooling actions are constrained by thermal states and the TES charge level, and UPS discharge can support IT execution while altering the grid power draw. A component-by-component assessment can therefore overestimate flexibility if it ignores recovery requirements, or underestimate it if it misses compensating actions across assets. Consequently, data centres can transition from being passive electrical loads to active assets that enhance the resilience and flexibility of the power system.

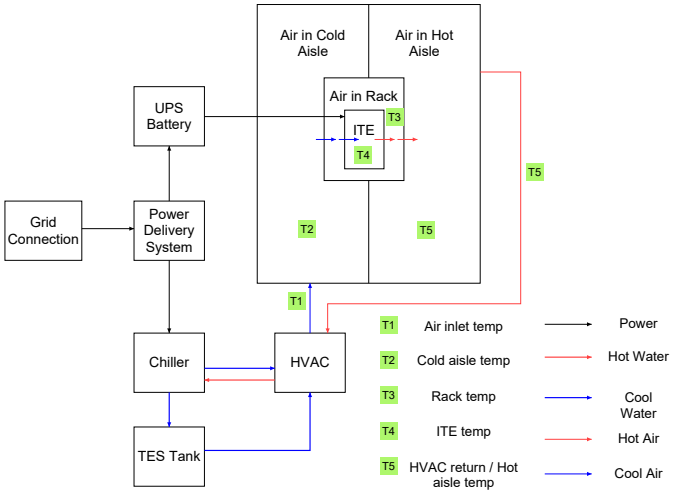


Fig. 1. Data centre layout and modelled temperature nodes. Redrawn using Python code prepared with assistance from ChatGPT Codex.

1.2 IT Workload Background

The fundamental role of a data centre is to execute IT workloads, which encompass all computational and data processing tasks. These workloads can be broadly classified based on their time sensitivity, as defined by service level agreements (SLAs) [11], [12]. Inflexible workloads, also known as interactive workloads, are latency-critical and require immediate execution to ensure quality of service for user-facing applications like real-time streaming or transaction processing.

In contrast, flexible IT workloads, often referred to as batch workloads, can tolerate execution delays within predefined time windows without impacting service quality. These typically non-user-facing tasks, such as periodic data analytics, backups, or machine learning model training, can be deferred or rescheduled. This ability to shift the timing of their execution is the primary source of IT-based flexibility, allowing the data centre to adjust its computational power draw in response to external signals like electricity prices or grid operational needs [12].

2. Literature Review

Data centre flexibility has attracted growing attention as an enabler of demand side participation in power systems, with a growing body of literature modelling flexibility at the asset level, predominantly focusing on IT workloads and, to a lesser extent, UPS batteries, cooling systems and other subsystems. A smaller body of work has attempted integrated models that capture the joint flexibility of multiple data centre assets. This review therefore first presents the flexibility modelling approaches found in the literature for each data centre asset and then discusses studies that move towards an integrated data centre flexibility assessment.

IT workload is the most fundamental component of a data centre and is the focus of many flexibility studies. One study develops a machine learning method to predict data centre energy consumption and server temperature [13]. These data are then used to optimise the scheduling of IT workloads to minimise fuel costs and power utilisation at times of high grid carbon intensity. On average, the implementation achieved a 6.5% load reduction for a winter day during periods of high CO_2 intensity. Radovanović et al. [14] performed predictive modelling of grid carbon intensity, trained day-ahead demand prediction models and used them to optimise IT workload for minimum carbon emissions. Cao et al. [12] propose and validate a data-driven methodology using real-world Alibaba trace data to assess the power flexibility potential of Internet Data Centres (IDCs) by rescheduling periodic IT workloads. Their findings quantify significant flexibility, particularly in early morning hours, revealing potential upward/downward load shifts of 400 kW and 325 kW, respectively—corresponding to approximately 50% and 40.6% of the facility's peak demand—by optimising job schedules based on power system needs and renewable energy availability. In this study, IT workloads are defined as either flexible or inflexible, with flexibility here referring to flexibility in time.

UPS batteries are currently providing power flexibility to the grid in numerous data centres around the world and have been shown to be successful [15]–[17]. Lithium-ion batteries, which are increasingly the dominant form used for UPS, are capable of rapidly providing or consuming power when needed [8]. These characteristics make them well suited to frequency regulation, a service many power providers require. The study in [8] found that UPS batteries could be used within their normal operating range to provide the required services to the grid and in doing so generate a reasonable revenue stream. Furthermore, the battery lifetime would not be affected, and the approach does not incur additional costs.

Cooling makes up a large proportion, as high as 40%, of data centre energy consumption [18]. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) recommends maintaining an indoor temperature of 18°C to 27°C to ensure optimal data centre performance [19]. Exploiting the thermal inertia provided by this temperature range has been shown to enable power consumption flexibility in data centres [20], [21]. Thermal energy storage (TES) systems can enhance cooling flexibility by decoupling the timing of heat removal from the cooling generation process. Du et al. [22] propose and evaluate a framework to enhance the energy flexibility of district heating (DH) systems integrated with data centre waste heat recovery. The study utilises dual short-term TES tanks to enable simultaneous peak shaving (demand-driven) and load shifting (price-driven) within the hybrid system. Simulation results demonstrated that this dual TES approach significantly improved flexibility, achieving up to a 10% peak demand reduction and a 2.1% load shift, leading to a 3.2% operational cost saving in the case study. A 'synergistic control strategy for data centre frequency regulation which uses both IT and cooling systems' has also been proposed [23]. The study found that a revenue saving of 4% could be made by implementing the proposed strategy. Once depleted, the thermal inertia of the data centre atmosphere / TES must be recovered to baseline levels. The recovery time taken to do so will affect how the flexibility of the asset can be utilised.

While individual assets offer flexibility, the flexibility characteristics change when these assets are integrated into one system. Maximising the overall potential requires coordinating multiple resources and evaluating their diverse characteristics, interactions, and constraints. One study proposes a co-optimisation of the IT workload and cooling system, recognising the strong operational link between them [23]. The work by Xiang et al. [24] identifies that data centre optimisation models often ignore IT equipment constraints like start-stop conditions and ramp rates, which can cause excessive wear. The authors propose a model integrating these constraints with facility thermal dynamics, solved using a Deep Deterministic Policy Gradient (DDPG) algorithm. This approach was then validated for practicality and efficiency using a large-scale simulation of a 100,000-device data centre. Xu et al. [25] propose an optimisation which integrates IT servers, cooling infrastructure, energy storage, generators and renewable energy, and aims to maximise power demand flexibility. The objective function is the cumulative change in energy taken from the grid before and after the optimisation was applied. Other studies use a similar approach, but instead minimise the difference between data centre power consumption and the desired power consumption of the power supply operator [26].

While individual data centre assets, including IT workloads, UPS batteries and cooling systems, have each been studied for their flexibility potential, the interplay between these assets and its impact on the overall flexibility of the data centre remain largely unexplored. Most existing studies adopt a facility-centric, cost-minimisation perspective and do not quantify the magnitude and duration of grid-facing flexibility that can be delivered once coupled asset constraints and post-event recovery are accounted for. As data centres scale towards the gigawatt level, this power-system-centric perspective becomes increasingly important, because system operators require duration-certified flexibility offers and a clear understanding of which assets enable them; this perspective remains largely absent from the literature.

Addressing these gaps, in this study, a flexibility analysis is performed to determine, for a given start time and

flexibility magnitude, a certified feasible duration for which this flexibility can be provided. This analysis is run for a large range of flexibility magnitudes and start times to establish a heatmap of flexibility potential. For each flexibility calculation, the contribution that each data centre asset makes is investigated. This analysis determines the contribution of each asset to the overall flexibility and reveals how the characteristics of each asset are exploited to maximise the flexibility potential. From a power-system-centric view, the magnitude and duration of flexibility quantify its utility. Thus, by quantifying these values, the benefit the modelled data centre can provide to the power system is determined.

3. Methodology

This section details the mathematical framework developed to quantify the demand side flexibility of a data centre by co-optimising its primary subsystems. The conceptual model of the data centre, illustrated in Figure 1, integrates three dynamically coupled components: the IT systems, which process both flexible and inflexible computational workloads; the power infrastructure, featuring an uninterruptible power supply which operates as an energy storage system (UPS ESS); and the cooling infrastructure, which includes a chiller coupled with a thermal energy storage (TES) tank and the CRAC system.

Our approach is centred on a comprehensive optimisation model that captures the operational dynamics and physical constraints of each subsystem. Flexibility is harnessed from three key sources: (1) shifting IT workloads in time, (2) strategically charging and discharging the UPS ESS, and (3) strategically managing the cooling system by utilising the TES and the inherent thermal inertia of the facility. By integrating these components into a unified framework, the data centre's total capacity to modulate its electricity consumption is evaluated, thereby quantifying its potential to provide valuable grid services.

The model is intentionally aggregate rather than a server-level or device-level digital twin. The objective of this work is to quantify the flexibility potential available to the power system from a data centre, not to reproduce the detailed operational behaviour of individual servers or cooling devices [27], [28]. The modelling therefore prioritises the state variables that determine flexibility feasibility: CPU utilisation and delay tolerance for IT work, UPS state of charge and converter limits for electrical storage, TES state of charge and facility temperatures for cooling recovery, and total grid power for service delivery. Likewise, the purpose of the cooling model is to capture the energy-flexibility interaction between thermal storage, cooling demand and electricity consumption, rather than detailed airflow dynamics within server racks. This abstraction preserves the operational couplings that matter for flexibility while keeping the repeated optimisation solves required by the flexibility assessment computationally tractable. The required computational work is preserved within the modelled execution windows. This maintains aggregate computational throughput under the adopted assumptions; application-specific service quality, task dependencies and server placement are not modelled.

Table II
Evidence Informing the Workload Assumptions

Evidence type	Sources	Use in this study
Observed utilisation ranges	[29], [30]	Context for CPU utilisation assumptions.
Delay-sensitivity classes	[31]	Mapping of workload classes into flexible and inflexible groups.
Batch workload flexibility	[12]	Flexible/inflexible workload energy shares and periodic workload shifting behaviour.
Flexibility case studies	[13], [23], [25], [32], [33]	Cross-checking plausible daily workload shapes and flexibility ranges.

3.1 IT Workload Methodology

In the literature, there is no consensus on IT workload characteristics and utilisation levels, as they vary significantly depending on the data centre size, type (e.g., enterprise, colocation, hyperscale) and workload nature (e.g., banking, AI). Consequently, even publicly available real-world datasets can differ substantially across facilities. Zaloumis [29] indicates that average CPU utilisation levels vary between 12% and 18% of total capacity, while Ghia [30] reports a daily average between 20% and 30%, noting that best practices can elevate this to 70%–80%.

An analysis of the Google cluster trace [31] classified data centre workloads into four categories based on their sensitivity to delay, labelled from "0" (least sensitive) to "3" (most sensitive). If categories 2 and 3 are considered inflexible, they account for approximately 30% of the total workloads, suggesting that up to 70% of the workloads could be flexible. Another study by Cao et al. [12], conducted using Alibaba's cluster trace, found that inflexible workloads represent 60% of all jobs but contribute only 30% of total energy consumption. In contrast, flexible workloads constitute 40% of jobs while consuming 70% of the energy. Furthermore, studies such as [13], [23], [25], [32], [33] reveal that there is no universally accepted ratio, and the estimated shares of flexible and inflexible IT workloads differ significantly across studies for a typical 24-hour period.

Based on the review of the literature, a dataset of reported workload ratios and graphs over a 24-hour time horizon was compiled. Table III retains the minimum, maximum and average values reported in that synthesis as context. The adopted workload ratios are modelling assumptions rather than measurements from a single facility. The "Proposed" columns show hourly means of the quarter-hour inputs actually used in the annual calculations; they should not be reconstructed from the contextual range columns. The evidence base informing these assumptions is consolidated in Table II. The retained range summaries do not establish a reproducible statistical derivation of each adopted input or a representative distribution across facilities. The exact 96 quarter-hour input rows are given in Supplementary Section S1.

Categorising the deferral flexibility of IT workloads, defined as the proportion of workloads that can be shifted and

Table III
IT WORKLOAD RATES

Time	Flexible Workload Ratio (%)				Inflexible Workload Ratio (%)				Total Workload Ratio (%)			
	Min	Max	Average	Proposed	Min	Max	Average	Proposed	Min	Max	Average	Proposed
00–01	17	75	37	37.00	20	39	32	26.50	55	80	68	63.50
01–02	13	72	36	31.75	25	38	32	22.00	46	66	56	53.75
02–03	18	70	38	29.25	25	37	32	16.25	43	50	47	45.50
03–04	15	68	37	24.50	19	38	26	13.00	34	37	36	37.50
04–05	17	64	36	27.00	8	36	21	6.75	25	37	31	33.75
05–06	15	59	32	23.25	2	39	21	8.00	22	36	29	31.25
06–07	14	51	29	20.00	13	40	25	15.00	20	36	28	35.00
07–08	24	42	31	24.00	18	39	27	21.50	24	52	38	45.50
08–09	23	35	30	25.50	30	53	41	27.50	31	65	48	53.00
09–10	27	49	35	24.25	36	65	49	35.50	38	85	62	59.75
10–11	23	50	35	17.50	37	62	51	38.50	45	87	66	56.00
11–12	19	50	35	17.50	42	70	56	40.50	50	92	71	58.00
12–13	17	49	33	21.50	40	51	47	39.00	52	89	71	60.50
13–14	16	49	33	24.75	36	60	49	35.50	54	85	70	60.25
14–15	16	52	35	27.00	35	68	52	33.50	55	87	71	60.50
15–16	17	49	35	24.00	33	70	53	34.75	57	82	70	58.75
16–17	26	43	37	27.50	35	62	52	35.00	63	78	71	62.50
17–18	37	39	38	41.50	20	60	39	26.75	67	75	71	68.25
18–19	32	49	39	45.00	10	63	37	26.00	70	73	72	71.00
19–20	23	54	35	45.50	9	67	38	25.50	62	77	70	71.00
20–21	22	58	36	45.50	8	62	37	26.50	63	83	73	72.00
21–22	21	61	38	41.00	7	61	37	29.25	63	87	75	70.25
22–23	17	64	38	40.50	6	55	35	26.00	61	89	75	66.50
23–24	16	69	40	42.00	2	45	29	21.00	56	80	68	63.00

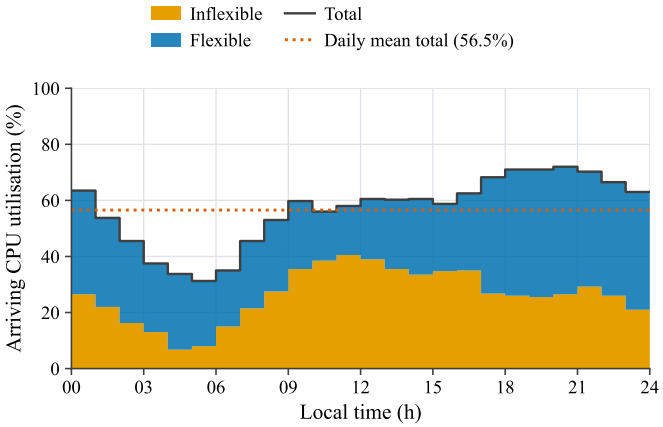


Fig. 2. Hourly means of the adopted quarter-hour arriving workload. The step outline shows total arriving CPU utilisation and the dotted line its daily mean. Plotted from the adopted inputs using Python code prepared with assistance from Claude Code.

the maximum duration by which they can be postponed, is essential for developing effective load shifting strategies. Drawing on insights from various studies [1], [12], a representative set of deferral windows applicable to flexible IT workloads over a 24-hour period is defined and summarised in Table IV.

Table IV presents hourly shares of flexible workloads across the four delay tranches: tranche $k = 1$ permits a final execution interval start 30 minutes after arrival, $k = 2$ 60 minutes, $k = 3$ 2 hours, and $k = 4$ 3 hours. The percentages represent the relative distribution within the flexible workload. Each hourly distribution is held constant over its four consecutive quarter-hour intervals, giving 96 input rows from the 24 hourly distributions. Figure 2 shows the flexible and inflexible arrival utilisation as arithmetic means of the four quarter-hour values within each labelled hour. Its average-total reference line is calculated from all 96 quarter-hour totals and is 56.5%. Together these profiles capture the time-dependent and duration-sensitive workload assumptions. A final eligible interval closes 15 minutes after its start: for example, a three-hour class arriving at noon may execute in the interval 15:00–15:15. Thus the class names denote deferral between interval starts, not continuous-time completion deadlines.

3.1.1 IT Power Consumption Model: In the literature, various approaches exist for modelling the power consumption of IT equipment, particularly servers. Given that server power consumption often constitutes the dominant portion of the total IT power draw, other components are frequently disregarded,

Table IV
HOURLY TRANCHE SHARES IN THE ADOPTED INPUTS (%)

Tranche	Latest interval start after arrival	Time																							
		00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$k = 1$	30 min	25	25	35	25	25	25	23	20	25	25	35	32	45	50	40	45	50	50	10	15	18	22	20	15
$k = 2$	60 min	25	25	17	15	20	15	28	20	15	17	22	20	25	20	20	25	15	20	20	20	12	18	16	20
$k = 3$	2 h	20	15	18	20	15	15	15	15	13	16	13	15	15	15	25	15	20	15	30	25	25	25	24	20
$k = 4$	3 h	30	35	30	40	40	45	34	45	47	42	30	33	15	15	15	15	15	15	40	40	45	35	40	45

and server power consumption is typically taken as a proxy for the overall IT power consumption. Numerous linear and non-linear models have been developed to represent server power consumption. A fundamental and widely accepted principle is that server power consumption is strongly correlated with its CPU utilisation. Based on this principle, Equation (1) was used in this study to estimate the electric power consumption associated with the IT workload [28], [34].

$$P^{\mathrm{IT}}(t) = P^{\mathrm{IT}}_{\mathrm{idle}} + (P^{\mathrm{IT}}_{\mathrm{max}} - P^{\mathrm{IT}}_{\mathrm{idle}}) \times u(t)^{1.32} \tag{1}$$

Therefore, the flexibility provided through IT workload shifting fundamentally depends on the rescheduling of CPU utilisation over time.

The core principle of workload shifting employed in this article relies on the concept of constant computational demand for each job. This demand, denoted as R, is defined by the required CPU resources and the duration for which they are needed. Representing computational demand as the product of CPU utilisation and execution duration is a common approximation in aggregate workload modelling, because the total amount of computational work can be interpreted as the accumulation of processor occupancy over time [12], [28], [31]. For example, a workload operating at 50% utilisation for two hours produces an equivalent computational demand to a workload operating at 100% utilisation for one hour. This abstraction is appropriate at the facility level considered here, where the modelling target is total processor occupancy over time rather than the instruction-level execution path of individual jobs. Execution can be at a lower or higher CPU utilisation for a longer or shorter duration, respectively. In all cases, the IT job must be fully executed within the maximum delay tolerance defined by its tranche. The crucial relationship of constant computational demand for a given IT job is mathematically represented in Equation (2) and conceptually illustrated in Figure 3.

$$R = u(t) \times \Delta T \tag{2}$$

As depicted in Figure 3, a job with a constant computational demand R can be executed over varying time frames ΔT by adjusting the allocated CPU utilisation $u(t)$. Decreasing CPU utilisation extends the job's duration, while increasing it shortens the duration, all while maintaining the same total computational demand (R). Consequently, the power consumption profile corresponding to the CPU utilisation changes accordingly. By strategically increasing or decreasing CPU utilisation in response to system requirements or grid signals, the associated power consumption can be modulated, enabling the provision of downward or upward flexibility, respectively.

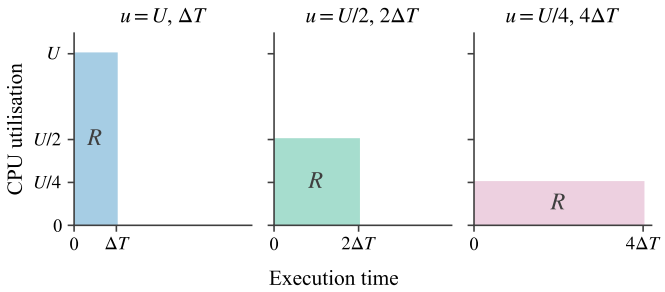


Fig. 3. Constant computational demand R executed at different CPU utilisations and durations. Drawn using Python code prepared with assistance from Claude Code.

3.1.2 IT Workload Formulation: The IT workload formulation presented in this paper accounts for both flexible and inflexible job types and incorporates the power consumption model formulated in Equation (1). The formulation is discretised into 15-minute time slots. Each annual solve covers a local day T_d and a 3 hour look-ahead, together denoted T_d^{ext}. For readability, T and T^{ext} below denote the current core and extended horizon. Work already unfinished at the opening boundary is included with its original arrival and deferral limit. New workload arrivals and full facility power are represented throughout the look-ahead.

The model conceptualises all servers of the entire data centre as a single representative server, treating all IT workloads as if processed by a single computational unit. This simplification enables a tractable yet representative analysis of IT workload deferral at the system level.

The flexible IT workloads are decomposed into tranches as shown in Table IV, with each tranche representing the portion of the jobs that can be shifted up to a specified maximum delay duration. The primary decision variable is $u(t, k, s)$, which denotes the CPU utilisation of tranche k, of a job originating at slot t, including carried arrivals and arrivals in the extended horizon, which is executed at time slot $s \in W_{t,k}$. IT workload shifting and execution are achieved while satisfying constraints (3)–(11).

The allocated CPU utilisation for any tranche must be non-negative and cannot exceed the maximum capacity, as shown in (4). The base IT power and energy consumption (i.e., without optimisation) are calculated as shown in (5) and (6),

$$P^{\mathrm{IT}}_{\mathrm{opt}}(s) = \widehat{f}\left(u^{\mathrm{inflex}}(s) + \sum_{(t,k)\in\mathcal{A}_d:\, s\in W_{t,k}} u(t,k,s)\right), \quad s \in T^{\mathrm{ext}}. \tag{3}$$

respectively. Without load shifting, the workload in the main 24 hour period consists of both inflexible jobs and flexible jobs that are executed at their time of arrival.

Under the load shifting scenario, the optimised power and energy consumption are calculated using (3) and (7). Full IT power is modelled in both the committed day and the look-ahead; baseline power is not subtracted from extension-period demand. This ensures that future work, cooling and storage interact through the same physical power balance. Annual settlement subsequently counts only committed intervals, as described in Section 4.2.1.

Let $\mathcal{A}_d$ denote the arrival/tranche pairs present in a horizon: unfinished cohorts carried from earlier days and new arrivals throughout T^{ext}_d. Their currently available CPU-hours are $A_{t,k,d}$, equal to original work $R_t\alpha_{t,k}$ for a new cohort and to its residual work for a carried cohort. Let $W_{t,k}$ contain the eligible intervals within the current horizon and L_d its final interval start. In (3), only eligible pairs contribute to execution at slot s. The same four-segment interpolation $\widehat{f}$ of Equation (1) is used in both scenarios. Its implementation is specified in Section 4.2.

IT jobs cannot be scheduled before their arrival or after their delay window, which is ensured by (8). While performing IT workload shifting, it is essential to ensure that all jobs are executed and completed within their maximum delay tolerance. These conditions are satisfied through (9) and (10). Constraint (9) ensures that the fractions $\alpha_{t,k}$ assigned to each tranche k of a job sum to one, distributing the entire job demand across its tranches. Constraint (10) requires completion when the final eligible interval lies within the horizon; otherwise it limits processing to the available work. Only execution in committed intervals reduces the work carried to the next day, so advisory look-ahead processing is never counted as completed work.

Finally, constraint (11) enforces per-time-slot CPU capacity. At each slot s, the CPU utilisation equals the inflexible load plus the contributions of all eligible flexible tranches. The summed utilisation is bounded by $u_{\max}$, ensuring no schedule exceeds the available CPU capacity at the time of execution.

$$0 \le u(t,k,s) \le u_{\max}, \quad (t,k) \in \mathcal{A}_d,\ s \in W_{t,k} \tag{4}$$

$$P^{\mathrm{IT}}_{\mathrm{base}}(s) = \widehat{f}\big(u^{\mathrm{inflex}}(s) + u^{\mathrm{flex,base}}(s)\big) \tag{5}$$

$$E^{\mathrm{IT}}_{\mathrm{base}}(t) = \Delta t \cdot P^{\mathrm{IT}}_{\mathrm{base}}(t), \quad \forall t \in T^{\mathrm{ext}} \tag{6}$$

$$E^{\mathrm{IT}}_{\mathrm{opt}}(t) = \Delta t \times P^{\mathrm{IT}}_{\mathrm{opt}}(t), \quad \forall t \in T^{\mathrm{ext}} \tag{7}$$

$$u(t,k,s) = 0 \quad \text{if } s < t \text{ or } s > t + D_k, \quad (t,k) \in \mathcal{A}_d \tag{8}$$

$$\sum_{k\in K} \alpha_{t,k} = 1 \quad \text{for each new arrival } t \tag{9}$$

$$\sum_{s\in W_{t,k}} u(t,k,s)\Delta t \begin{cases} = A_{t,k,d}, & t + D_k \le L_d, \\ \le A_{t,k,d}, & t + D_k > L_d, \end{cases} \tag{10}$$

$$u^{\mathrm{inflex}}(s) + \sum_{(t,k)\in\mathcal{A}_d:\, s\in W_{t,k}} u(t,k,s) \le u_{\max}, \quad s \in T^{\mathrm{ext}}. \tag{11}$$

3.2 UPS ESS Mathematical Model and Constraints

The key decision variables in this model are the state of charge ($E^{\mathrm{UPS}}(t)$), the charging and discharging powers ($P^{\mathrm{UPS}}_{\mathrm{ch}}(t)$, $P^{\mathrm{UPS}}_{\mathrm{disch}}(t)$), and their corresponding binary status indicators ($z^{\mathrm{UPS}}_{\mathrm{ch}}(t)$, $z^{\mathrm{UPS}}_{\mathrm{disch}}(t)$). The dynamic operation of a UPS ESS, transitioning from time t to $t + \Delta t$, is governed by the set of equations and inequalities shown in (12)–(19).

The energy level of the batteries at the end of time interval t is calculated based on the charging and discharging power applied during that interval, as formulated in Equation (12) [35]. The energy level of the batteries must be maintained within their predefined operational boundaries as shown in (13) [35]. The SoC limits prevent overcharging and deep discharging, as illustrated in (14). Furthermore, the opening energy of each day equals the preceding committed closing energy, as specified in (15). Initial UPS energy is 600 kWh and the operational lower bound retains a 50% reserve. No daily cyclic recharge is imposed.

The charging and discharging powers are constrained within the allowable limits of the UPS ESS, as shown in (16) and (17). Equation (18) guarantees that charging and discharging do not occur simultaneously [35]. Net power, as shown in (19), is defined as the difference between charging and discharging power, and is a single, signed variable representing the total instantaneous power exchange in the UPS. This allows for a physically accurate constraint on the UPS converter, whether it is charging from the grid or powering the IT equipment to reduce the grid load. Discharge cannot exceed IT demand, and the facility does not export electricity [36].

This constraint model effectively captures the physical limitations of the power converter. If the battery is charging, then $P^{\mathrm{UPS}}_{\mathrm{disch}}(t) = 0$ and $P^{\mathrm{UPS}}_{\mathrm{net}}(t) = P^{\mathrm{UPS}}_{\mathrm{ch}}(t)$. If it is discharging, then $P^{\mathrm{UPS}}_{\mathrm{ch}}(t) = 0$ and $P^{\mathrm{UPS}}_{\mathrm{net}}(t) = -P^{\mathrm{UPS}}_{\mathrm{disch}}(t)$. If idle, both powers are zero.

3.3 Cooling System Model

This section models data centre cooling infrastructure to assess its flexibility potential. The methodology is based on the validated thermodynamic model by Cupelli et al. [37], which is adapted in this work by incorporating a thermal energy storage (TES) tank and scaling the parameters for a 1 MW IT capacity data centre. Although modern deployments increasingly reach the 100 MW scale, the 1 MW case is retained as a normalised unit that keeps the coupled

$$E^{\mathrm{UPS}}(t+\Delta t) = E^{\mathrm{UPS}}(t) + (\eta_{\mathrm{ch}}^{\mathrm{UPS}} \cdot P_{\mathrm{ch}}^{\mathrm{UPS}}(t) \cdot \Delta t) - \left(\frac{P_{\mathrm{disch}}^{\mathrm{UPS}}(t)}{\eta_{\mathrm{disch}}^{\mathrm{UPS}}} \cdot \Delta t\right), \quad \forall t \in T^{\mathrm{ext}} \tag{12}$$

$$E_{\mathrm{min}}^{\mathrm{UPS}} \leq E^{\mathrm{UPS}}(t) \leq E_{\mathrm{max}}^{\mathrm{UPS}}, \quad \forall t \in T^{\mathrm{ext}} \tag{13}$$

$$E_{\mathrm{min}}^{\mathrm{UPS}} = SoC_{\mathrm{min}} \cdot E_{\mathrm{base}}^{\mathrm{UPS}}, \quad E_{\mathrm{max}}^{\mathrm{UPS}} = SoC_{\mathrm{max}} \cdot E_{\mathrm{base}}^{\mathrm{UPS}} \tag{14}$$

$$E_{d+1,\mathrm{start}}^{\mathrm{UPS}} = E_{d,\mathrm{core\ end}}^{\mathrm{UPS}}, \quad E_{1,\mathrm{start}}^{\mathrm{UPS}} = E_{\mathrm{base}}^{\mathrm{UPS}}. \tag{15}$$

$$z_{\mathrm{ch}}^{\mathrm{UPS}}(t) \cdot P_{\mathrm{ch,min}}^{\mathrm{UPS}} \leq P_{\mathrm{ch}}^{\mathrm{UPS}}(t) \leq z_{\mathrm{ch}}^{\mathrm{UPS}}(t) \cdot P_{\mathrm{ch,max}}^{\mathrm{UPS}}, \quad \forall t \in T^{\mathrm{ext}} \tag{16}$$

$$z_{\mathrm{disch}}^{\mathrm{UPS}}(t) \cdot P_{\mathrm{disch,min}}^{\mathrm{UPS}} \leq P_{\mathrm{disch}}^{\mathrm{UPS}}(t) \leq z_{\mathrm{disch}}^{\mathrm{UPS}}(t) \cdot P_{\mathrm{disch,max}}^{\mathrm{UPS}}, \quad \forall t \in T^{\mathrm{ext}} \tag{17}$$

$$z_{\mathrm{ch}}^{\mathrm{UPS}}(t) + z_{\mathrm{disch}}^{\mathrm{UPS}}(t) \leq 1, \quad \forall t \in T^{\mathrm{ext}} \tag{18}$$

$$P_{\mathrm{net}}^{\mathrm{UPS}}(t) = P_{\mathrm{ch}}^{\mathrm{UPS}}(t) - P_{\mathrm{disch}}^{\mathrm{UPS}}(t), \quad \forall t \in T^{\mathrm{ext}} \tag{19}$$

optimisation transparent and allows the flexibility results to be interpreted per MW of IT capacity; transfer to a larger facility remains conditional on its workload, equipment ratios and operating limits. Flexibility is harnessed from two primary sources. The first is the TES tank, which provides downward flexibility by charging (increasing electrical load) and upward flexibility by discharging (allowing the chiller to reduce its load). The second is the inherent thermal mass of the data centre components, such as the IT servers, racks, and air in the data centre. By allowing temperatures to fluctuate within their component-specific bounds (cold aisle 18 –23°C), the masses act as a thermal buffer. The thermal buffer allows the chiller's power consumption to be modulated within the modelled IT operating limits. Gross cooling is additionally constrained to be at least IT heat generation; this adopted rule limits the amount of flexibility credited to thermal accumulation. The cooling system of the data centre is considered a closed-loop thermodynamic system where a central chiller provides cooled water to a CRAC unit, either directly or via the TES tank. This air is circulated through the data centre to remove heat generated by the IT equipment, with a constant air mass flow rate, $\dot{m}_a$. The chiller is considered the main variable electrical load, while the power consumption of auxiliary components is treated as the constant overhead $P_{\mathrm{OD}}^{\mathrm{Grid}}$. The governing equations describing the system's thermal dynamics and operational constraints are presented below.

Equation (20) represents the overall thermal energy balance, where the rate of change of energy stored in the data centre's thermal masses (aisles, racks, ITE) equals the net heat flow from the environment, ITE, and cooling system, where $Q_{\mathrm{out}}(t) = G_{cd}(T_{\mathrm{out}} - T_{CA}(t+1))$. It is assumed that all ITE electrical power converts to heat, i.e., $P^{\mathrm{IT}}(t) = Q^{\mathrm{IT}}(t)$. The discrete-time temperature dynamics for the supply air, ITE, racks, cold aisle, and hot aisle are governed by Equations (26) through (30), employing an implicit backward Euler method. Here $t+1$ denotes the boundary after interval t (the same boundary as $t+\Delta t$ in the UPS equation). In Equation (20), $Q_j(t) = C_j[T_j(t+1) - T_j(t)]/(3600\Delta t)$ and the effective cooling of the represented nodes is κQ_{cool}, consistent with the calibrated airflow factor in the state equations.

The primary distinction between the equations for cold aisle temperature (29) and hot aisle temperature (30) stems from the modelled hot aisle containment, which prevents any heat loss from the hot aisle to the external environment. Equation (21) shows the total cooling power is the sum of contributions from the CRAC and the discharging TES. Equations (22) and (23) relate the chiller's electrical power consumption to the thermal cooling delivered, governed by its coefficient of performance ($\mathrm{COP}_{\mathrm{chiller}}$). The chiller's total power draw is capped by constraint (24). The cooling power is limited by (31) to prevent overcooling the cold aisle. The TES state of charge is updated by Equation (25), accounting for charging/discharging efficiencies. The operational limits of the TES are defined in (32) through (34), which set maximum power levels and prevent simultaneous charging and discharging. The TES energy level is constrained by (36), while (35) passes the committed closing state to the next day, without requiring a daily return to the initial state. To ensure safe operation, constraint (37) enforces the upper and lower temperature bounds for all thermal components, including the final state boundary. Initial IT, rack, cold aisle and hot aisle temperatures are $(28.5, 26, 20, 27)$°C; subsequent daily openings inherit all four states. Storage energy bounds also apply at the final boundary.

4. Case Studies: Integrated Data Centre Model

This section evaluates the integrated operation of a data centre in which IT workload scheduling, UPS ESS dispatch, and cooling/TES control are co-optimised. Building on the component models in Section 3, a whole-facility formulation is assembled, linked by electric and thermal coupling constraints, and tested under realistic price signals. The scenarios are designed to answer two questions: (i) using cost-optimised operation as a baseline, how long can a requested upward or downward power deviation be sustained, for any given start time, while meeting the modelled workload and operating constraints; and (ii) to what extent can an integrated optimum energy management algorithm reduce annual electricity cost? This two-step approach models a realistic scenario where an operator first determines their daily operating schedule and then assesses the remaining capacity for flexibility services. The annual analysis measures electricity cost savings; additional service revenue is not calculated. The overall workflow linking the three scenarios, their inputs and their outputs is summarised in Figure 4.

A hypothetical data centre with 1 MW IT capacity and associated infrastructure is considered. Annual electricity prices are the 2025 Great Britain day-ahead Intermittent Market Reference Prices (IMRP) published by the Low Carbon Contracts Company [38]. Each hourly price is applied to four quarter-hour intervals, retaining negative values. Table V gives prices for the selected illustrative day, 22 September 2025; annual costs use all 365 days. The final look-ahead uses actual 1 January 2026 prices rather than repeating an earlier day's

$$Q_{CA}(t) + Q_{HA}(t) + Q_R(t) + Q_{ITm}(t) = Q_{out}(t) + Q^{\mathrm{IT}}(t) - \kappa Q_{\mathrm{cool}}(t), \quad t \in T^{\mathrm{ext}} \tag{20}$$
$$Q_{\mathrm{cool}}(t) = Q^{\text{Chil-CRAC}}(t) + Q^{\text{TES-CRAC}}(t), \quad \forall t \in T^{\mathrm{ext}} \tag{21}$$
$$Q^{\text{Chil-CRAC}}(t) = P^{\text{Chil-CRAC}}(t) \cdot \mathrm{COP}_{\mathrm{chiller}}, \quad \forall t \in T^{\mathrm{ext}} \tag{22}$$
$$Q^{\text{Chil-TES}}(t) = P^{\text{Chil-TES}}(t) \cdot \mathrm{COP}_{\mathrm{chiller}}, \quad \forall t \in T^{\mathrm{ext}} \tag{23}$$
$$P^{\text{Chil-CRAC}}(t) + P^{\text{Chil-TES}}(t) \le P^{\mathrm{Chiller}}_{\max}, \quad \forall t \in T^{\mathrm{ext}} \tag{24}$$
$$E^{\mathrm{TES}}(t+1) = E^{\mathrm{TES}}(t) + [\eta^{\mathrm{TES}}_{\mathrm{ch}} Q^{\text{Chil-TES}}(t) - Q^{\text{TES-CRAC}}(t)/\eta^{\mathrm{TES}}_{\mathrm{dis}}]\Delta t, \quad t \in T^{\mathrm{ext}} \tag{25}$$
$$T_{Ain}(t) = T_{HA}(t+1) - Q_{\mathrm{cool}}(t)/(\dot{m}_a c_{pa}), \quad t \in T^{\mathrm{ext}} \tag{26}$$
$$T_{IT}(t+1) = T_{IT}(t) + \frac{3600\Delta t}{C_{IT}}[P^{\mathrm{IT}}(t) - G_{cv}(T_{IT}(t+1) - T_R(t+1))], \quad t \in T^{\mathrm{ext}} \tag{27}$$
$$T_R(t+1) = T_R(t) + \frac{3600\Delta t}{C_R}[\dot{m}_a \kappa c_{pa}(T_{CA}(t+1) - T_R(t+1)) + G_{cv}(T_{IT}(t+1) - T_R(t+1))], \quad t \in T^{\mathrm{ext}} \tag{28}$$
$$T_{CA}(t+1) = T_{CA}(t) + \frac{3600\Delta t}{C_{CA}}[\dot{m}_a \kappa c_{pa}(T_{Ain}(t) - T_{CA}(t+1)) - G_{cd}(T_{CA}(t+1) - T_{out})], \quad t \in T^{\mathrm{ext}} \tag{29}$$
$$T_{HA}(t+1) = T_{HA}(t) + \frac{3600\Delta t}{C_{HA}}[\dot{m}_a \kappa c_{pa}(T_R(t+1) - T_{HA}(t+1))], \quad t \in T^{\mathrm{ext}} \tag{30}$$
$$P^{\mathrm{IT}}(t) \le Q_{\mathrm{cool}}(t) \le (T_{HA}(t+1) - T_{CA,min})\dot{m}_a c_{pa}, \quad t \in T^{\mathrm{ext}} \tag{31}$$
$$0 \le Q^{\text{Chil-TES}}(t) \le z^{\text{Chil-TES}}(t) \cdot Q^{\text{Chil-TES}}_{\max}, \quad \forall t \in T^{\mathrm{ext}} \tag{32}$$
$$0 \le Q^{\text{TES-CRAC}}(t) \le z^{\text{TES-CRAC}}(t) \cdot Q^{\text{TES-CRAC}}_{\max}, \quad \forall t \in T^{\mathrm{ext}} \tag{33}$$
$$z^{\text{Chil-TES}}(t) + z^{\text{TES-CRAC}}(t) \le 1, \quad \forall t \in T^{\mathrm{ext}} \tag{34}$$
$$E^{\mathrm{TES}}_{d+1,\mathrm{start}} = E^{\mathrm{TES}}_{d,\mathrm{core\ end}}, \quad E^{\mathrm{TES}}_{1,\mathrm{start}} = 0.5E^{\mathrm{TES}}_{\max} \tag{35}$$
$$0 \le E^{\mathrm{TES}}(t) \le E^{\mathrm{TES}}_{\max}, \quad \forall t \in T^{\mathrm{ext}} \tag{36}$$
$$T_{j,min} \le T_j(t) \le T_{j,max}, \qquad \forall j \in \{\mathrm{Ain, CA, R, IT, HA}\}, \quad \forall t \in T^{\mathrm{ext}} \tag{37}$$

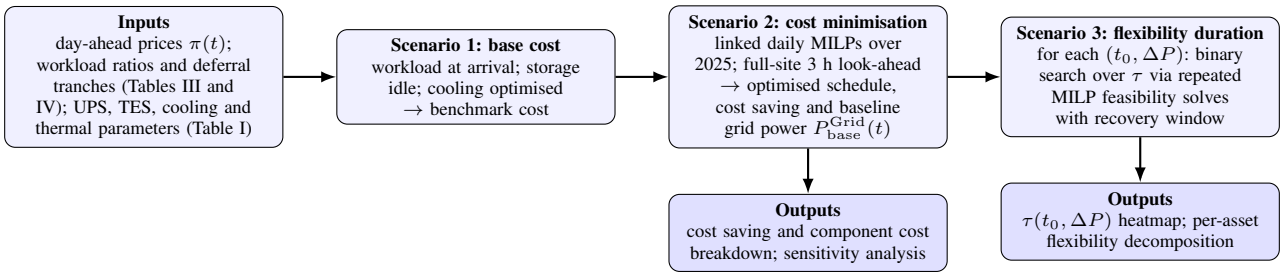


Fig. 4. Workflow of the three-scenario optimisation and flexibility assessment. Scenario 1 provides the workload-at-arrival cost benchmark, Scenario 2 establishes the cost-optimised baseline, and Scenario 3 identifies certified feasible flexibility durations for each requested start time and power deviation while enforcing recovery. Diagram code prepared with assistance from ChatGPT Codex.

Table V
DAY-AHEAD REFERENCE PRICES ON 22 SEPTEMBER 2025

Hour	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Price (GBP/MWh)	60.37	58.18	55.13	53.29	54.32	68.13	94.75	105.38	103.19	87.89	79.76	71.95	69.94	65.74	71.41	73.49	81.35	100.79	125.83	124.39	110.42	88.80	83.04	82.69

prices. All principal parameter values remain detailed in Table I. The 1 MW value is the IT rating, while the structural grid import bound is 1,723.095 kW including cooling, UPS charging and auxiliaries. Workload profiles repeat by local clock time, with fixed ambient temperature and COP.

4.1 Scenario 1: Base Cost Calculation

In Scenario 1, all workload executes at arrival and the UPS and TES remain idle. Cooling is scheduled to minimise electricity cost while maintaining the prescribed temperature limits. The cold aisle consequently operates predominantly at its upper limit of 23°C, with occasional cooling in advance of higher electricity prices. The purpose of this scenario is to define a base case that provides a benchmark for measuring the potential savings achieved through demand side flexibility. The total data centre electricity cost is defined as:

$$\min \sum_{t\in T^{\text{ext}}} \left(P_{\text{IT}}^{\text{Grid}}(t) + P_{\text{OD}}^{\text{Grid}} + P_{\text{ch}}^{\text{UPS}}(t) + P^{\text{Chil-CRAC}}(t) + P^{\text{Chil-TES}}(t)\right) \cdot \Delta t \cdot \pi(t)/1000, \tag{40}$$

$$P_{\text{opt}}^{\text{IT}}(t) = P_{\text{IT}}^{\text{Grid}}(t) + P_{\text{disch}}^{\text{UPS}}(t), \quad \forall t \in T^{\text{ext}} \tag{41}$$

$$0 \le P_{\text{IT}}^{\text{Grid}}(t), P_{\text{OD}}^{\text{Grid}}, P_{\text{ch}}^{\text{UPS}}(t), P^{\text{Chil-CRAC}}(t), P^{\text{Chil-TES}}(t), \quad \forall t \in T^{\text{ext}} \tag{42}$$

$$\sum_{t\in T} \left(P_{\text{IT}}^{\text{Grid}}(t) + P_{\text{OD}}^{\text{Grid}} + P^{\text{Chil-CRAC}}(t)\right) \cdot \Delta t \cdot \pi(t)/1000, \tag{38}$$

$$P_{\text{OD}}^{\text{Grid}} = \gamma_{\text{OD}} \cdot \overline{P}_{\text{IT+cool}}^{\text{ref}} = 0.07 \times 758.5 = 53.095 \text{ kW} \tag{39}$$

where $\pi(t)$ is the signed day-ahead reference price at time (t) in GBP/MWh; the factor $1/1000$ converts kW to MW. The expression is settled over each committed day and summed over 2025. The UPS and TES tank are not utilised and so incur no dispatch electricity cost in this scenario. Auxiliary electricity demand is assumed to be 7% of a reference mean IT and cooling load of 758.5 kW, giving a constant auxiliary load of 53.095 kW. This value is applied in all scenarios. Both scenarios use the same thermal limits, so the comparison measures the additional savings from workload scheduling and UPS and TES operation.

4.2 Scenario 2: Cost Minimisation

This scenario calculates the cost-optimised operating conditions by shifting IT workloads over time, utilising the UPS as an energy storage system (ESS), and leveraging the thermal mass of the data centre and TES tank for energy storage. This optimisation is highly dependent on the electricity pricing mechanism. Sensitivity to the flexible workload share and UPS and TES capacities is examined in Section 5.4. The objective function is to minimise the data centre's total electricity cost, as defined in Equation (40). This is subject to some additional constraints presented in Equations (41)–(42). Equation (41) ensures that the power supplied to the IT equipment from the grid and the UPS sums to the optimised IT power demand ($P_{\text{opt}}^{\text{IT}}(t)$), which is defined in Equation (3). Power from auxiliary devices, $P_{\text{OD}}^{\text{Grid}}$, is kept constant at the value defined in Scenario 1, so it is included in the objective but is not an optimisation decision. Equation (42) enforces the non-negativity of all power variables.

Note that the objective explicitly includes grid withdrawals (e.g. $P_{\text{IT}}^{\text{Grid}}(t)$) and charging powers (e.g. $P_{\text{ch}}^{\text{UPS}}(t)$, $P^{\text{Chil-TES}}(t)$). UPS discharge $P_{\text{disch}}^{\text{UPS}}(t)$ therefore does not appear directly in the cost sum; its economic effect is realised implicitly through Equation (41), which reduces $P_{\text{IT}}^{\text{Grid}}(t)$ when the UPS discharges.

The resulting optimisation problem is a mixed-integer linear programme (MILP), as it includes binary variables for the UPS charging/discharging status and TES operation alongside continuous power, energy, temperature and CPU utilisation decision variables. In this scenario, the integrated data centre model is used in linked daily MILP optimisations over the full year. To maintain the model's linearity, the non-linear function in Equation (1) is linearised using a piecewise linear approximation. This technique approximates the non-linear curve with a series of connected line segments. The implemented four segments have breakpoints $b_i = (i/4)^{1.5}$, $i = 0, \ldots, 4$, and use Pyomo's DLOG representation; the maximum approximation error is 6.241 kW (0.624% of rated IT power). Scenario 1 evaluates the identical interpolation at its fixed utilisation. The model was implemented in Python using the Pyomo modelling library and solved using HiGHS. The retained run record identifies Python 3.12.0, Pyomo 6.10.1 and HiGHS 1.15.1 on Windows 11 with an AMD Ryzen 7 9800X3D. The standard daily time limit is 300 s, and the requested relative MILP gap is 0.1%; feasible time-limited solutions and diagnostic checks are disclosed in Supplementary Section S5. These linked solves are not a single globally optimal year-long programme.

4.2.1 Annual operation and cost accounting: Scenarios 1 and 2 are evaluated over all 365 local days of 2025. Each solve contains the local-day core T_d and twelve look-ahead intervals, with full workload arrivals, cooling and storage operation throughout. Only core decisions are committed. The next day inherits UPS and TES energy, all four temperatures and each unfinished arrival/tranche pair with its original execution window. Look-ahead controls are reoptimised; their trial execution is not counted as completed work.

For a cohort available in day d, its committed residual is

$$A_{t,k,d+1} = A_{t,k,d} - \sum_{s\in T_d \cap W_{t,k}} u(t,k,s)\Delta t, \tag{43}$$

provided it has arrived before the next core boundary. New look-ahead arrivals are generated afresh when their arrival day is processed. Physical time follows a continuous UTC timeline; local-day and workload labels use Europe/London. Normal days have 96 committed intervals, with 92 and 100 on daylight-saving transitions. The resulting year contains 35,040 quarter-hours.

Annual settlement counts each committed interval once:

$$C_{\text{year}}^{(i)} = \sum_{d=1}^{365} \sum_{t\in T_d} P^{\text{Grid},(i)}(t)\pi(t)\Delta t/1000, \tag{44}$$

where $i = 1, 2$. The percentage saving is $100(C_{\text{year}}^{(1)} - C_{\text{year}}^{(2)})/C_{\text{year}}^{(1)}$. Overlapping look-ahead costs are excluded until their intervals become committed. The final optimisation extends into the first three hours of 1 January 2026 using observed prices and the adopted demand profile. These intervals inform decisions near the year boundary but are excluded from reported 2025 electricity costs. No monetary

value is assigned to energy remaining in storage at the end of the optimisation horizon. Supplementary Section S2 details the procedure and year-end workload treatment, and Section S5 records existing terminal and solver checks.

This retrospective annual assessment uses known inputs within each daily horizon. It avoids extrapolating a single day's savings, while leaving forecast uncertainty, network tariffs, battery degradation and flexibility-service revenues outside the objective.

4.3 Scenario 3: Flexibility Duration Calculation

The third scenario evaluates the data centre's capacity to provide flexibility in response to a direct grid signal. This approach models a realistic situation where the grid operator requests a specific magnitude of power change for a certain duration, for example during planned balancing events or unforeseen emergencies. The optimised baseline from Scenario 2 is taken and from it a certified feasible duration for which the data centre can sustain specified levels of flexibility magnitude is determined. This is calculated at any given time slot over the 24-hour period. Here, flexibility magnitude, ΔP (kW), is defined as the power deviation from the optimised baseline, with upward flexibility (load reduction) represented by negative magnitudes and downward flexibility (load increase) by positive magnitudes. For each combination of start time t_0 and requested flexibility magnitude ΔP, a continuous period τ is identified over which the system can track a correspondingly modified grid power trajectory while maintaining operational feasibility and respecting all physical and thermal constraints. This temporal measure of flexibility provides a key input for assessing the data centre's potential participation in demand response schemes and ancillary services markets by quantifying its ability to meet specific grid needs.

The flexibility analysis takes the optimised baseline from Scenario 2 as the reference trajectory and fixes the initial states at the chosen start time t_0. Each IT job may already have been shifted by a variable amount, so its remaining flexibility is its original execution window less the time already elapsed. The original four classes are retained, but unfinished work is tracked for each arrival/tranche pair rather than aggregated into new delay classes. Equations (45) and (46) express the remaining eligible execution window. This prevents a job that has already been deferred from receiving a fresh full deferral allowance.

Another constraint added to the formulation described in Section 3 is the core requirement to deviate the power drawn from the grid, relative to the Scenario 2 baseline, at the requested level over the chosen window. Denoting the baseline grid power draw as $P^{\mathrm{Grid}}_{\mathrm{base}}(t)$ and the new grid power draw as $P^{\mathrm{Grid}}(t)$, the flexibility constraint is given in (47).

$$K = \{1, 2, 3, 4\}, \quad (t, k) \in \mathcal{A}_{t_0} \tag{45}$$

$$W^{\mathrm{event}}_{t,k} = \{s : \max(t, t_0) \le s \le t + D_k\} \cap T^{\mathrm{event}} \tag{46}$$

$$\left| P^{\mathrm{Grid}}(t) - P^{\mathrm{Grid}}_{\mathrm{base}}(t) - \Delta P \right| \le P_{\mathrm{tol}}, \quad t_0 \le t < t_0 + \tau. \tag{47}$$

For each start time t_0 and flexibility magnitude ΔP, the feasibility of providing the given ΔP is tested over a series of durations. The optimisation described in Scenario 2 is run for each duration over the event and a further twelve-slot (three-hour) recovery window. The flexibility constraint in Equation (47) applies only during service delivery. During recovery, all physical, thermal, storage and workload constraints continue to apply but the grid power deviation is no longer enforced. At the recovery boundary $H = t_0 + \tau + 3$ h, UPS and TES energy must be at least their undisturbed Scenario 2 values, each of the four temperatures must be within 0.05°C of its corresponding value, and every cohort's unfinished work must be no greater than its undisturbed residual plus 10^{-7} CPU-hours. During recovery, unfinished flexible work may be rescheduled within its remaining original execution window; its deadline is unchanged. These comparative conditions prevent additional depletion or unfinished work from being left beyond recovery.

If an audited feasible solution is found for a given duration τ, the data centre can provide that level of flexibility for the given duration. The same feasibility test is then repeated for a longer duration; shorter candidates are explored after unsuccessful trials. The duration is modified using a logarithmic search, with an adjacent-duration check. Candidate durations are multiples of 15 minutes and capped at twelve hours or the remaining local day. An unresolved solve is not proof of infeasibility. Moreover, changing τ changes the recovery boundary and its reference states, so adjacent checks do not prove that all longer durations are infeasible. Each reported duration has a solution that satisfies the operating and recovery constraints. Results that reach the assessment limit or have an unresolved next trial are identified separately. The search does not establish the maximum possible duration for every request. Supplementary Section S3 gives the full procedure and status conventions.

22 September 2025 was selected because its twelve standardised indicators of prices and facility operation were closest to their annual means among eligible days. Supplementary Section S3 lists the indicators and eligibility criteria. Starts are sampled hourly; requests are -100 to -500 kW in 50 kW steps and $+25$, $+50$ and $+75$ kW, giving 288 combinations. Each event inherits the actual coordinated state and cohort ledger, without preconditioning. All scenarios retain the same 18– 23°C cold aisle bounds. These results show how flexibility varies with operating conditions on the selected day; they do not establish how frequently the same service could be delivered throughout the year.

4.4 Sensitivity Analysis Design

To assess the dependence of the cost-optimisation results on resource assumptions, a one-at-a-time (OAT) sensitivity analysis is performed around the base case, covering the IT load flexibility ratio and equipment sizing. The flexible workload share is scaled by moving load between the flexible and inflexible pools while preserving the total CPU demand in every time slot, so that the flexibility share is isolated from the

demand level; at high multipliers the share saturates in slots that are already almost fully flexible. The TES and UPS energy capacities are scaled while keeping their charge/discharge power limits unchanged, isolating the sizing of the energy buffers. Multipliers are 0.5, 0.75, 1.25 and 1.5, with the shared central case completing each curve. All cases cover 365 days and use the same workload-at-arrival annual comparator. Storage initial energy fractions and the 50% UPS reserve fraction are preserved as capacities change. All resource sensitivities use the same observed 2025 price series. A separate controlled price-volatility sweep is not included; the annual evaluation captures realised price variation but does not isolate its causal effect. The results are presented in Section 5.4. Additional UPS electricity savings are used to estimate the maximum incremental investment supported under assumed discount rates and battery lifetimes. Supplementary Section S6 gives the assumptions and calculations.

5. Results

5.1 *Scenarios 1 and 2: Operating Profiles*

On 22 September, coordinated operation shifts some electricity consumption away from expensive periods through workload scheduling and storage operation (Figure 5). Workload scheduling also changes the cooling requirement. The resulting profile depends on both electricity prices and the workload and storage states inherited from preceding intervals. Work is deferred only within its permitted execution window, and the annual cost saving cannot be inferred from this selected day.

Figure 6 provides a detailed decomposition of the data centre's power consumption and resource dispatch strategy under the cost-minimisation objective of Scenario 2. The chart visualises the co-optimised management of various electrical loads and energy storage assets over the 24 hour period. The positive stacked areas represent the components of power drawn from the grid, including IT, CRAC, and charging power for the energy storage systems. The negative stacked areas indicate UPS discharge supplying IT demand. TES discharge reduces the direct chiller electricity needed for cooling; it is not a separate electrical injection. The dashed black line shows the Scenario 1 benchmark grid import, illustrating the impact of the flexibility measures.

When electricity prices are favourable, the optimisation algorithm can draw more power from the grid than required in the base case. This surplus energy can charge the TES tank, effectively storing cooling produced with cheaper electricity, or charge the UPS. It can also be used to process eligible IT workloads, with a consequent change in CRAC power consumption. Conversely, grid power can be reduced by dispatching the TES tank and UPS and by shifting IT workload as shown in Figure 5(b). These decisions establish the operating baseline from which the following flexibility requests are assessed; their annual economic effect is reported after the physical flexibility results.

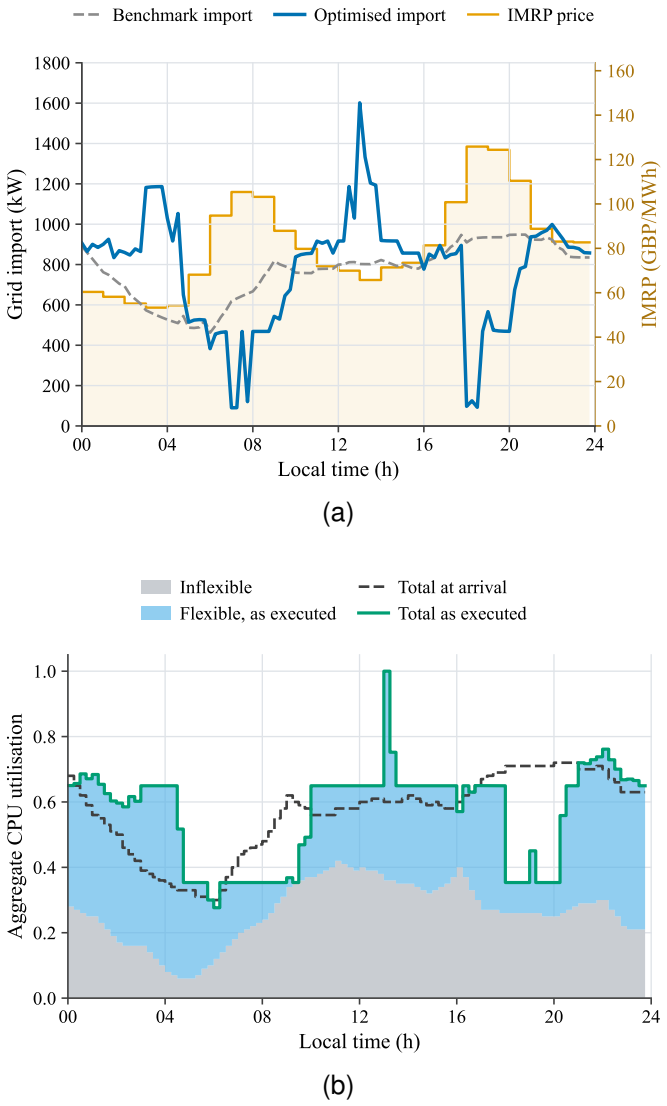


Fig. 5. (a) Grid power and reference prices on 22 September 2025. (b) IT workload arrival and execution: the dashed line shows total arrivals, the grey and blue areas show inflexible and executed flexible workload, and the green line shows total execution. Scenario 2 supplies the undisturbed trajectory from which flexibility is provided in Scenario 3. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

5.2 *Scenario 3 Results*

Scenario 3 proposes a comprehensive test to identify certified feasible flexibility durations for a set of flexibility magnitudes and start times. The resulting data are three-dimensional and can be effectively visualised as a heatmap. This heatmap is shown in Figure 7, with τ represented by the colour and (t_0, ΔP) shown on the (x, y) axes. Each cell corresponds to one evaluated $(t_0, \Delta P)$ pair, with the printed value giving a certified feasible duration τ in hours. The underlying model operates at a 15-minute resolution; the assessed start times are sampled hourly across the operating day. The top three rows (positive ΔP) correspond to downward flexibility and the remaining rows (negative ΔP) to upward flexibility.

Figure 7 illustrates significant variations in flexibility duration τ with start time t_0 and deviation ΔP. A 100 kW upward request can be sustained for 1 h from 06:00 and 6 h from noon; a 150 kW reduction lasts 5 h from 13:00. A 25 kW downward request reaches the twelve-hour assessment cap at

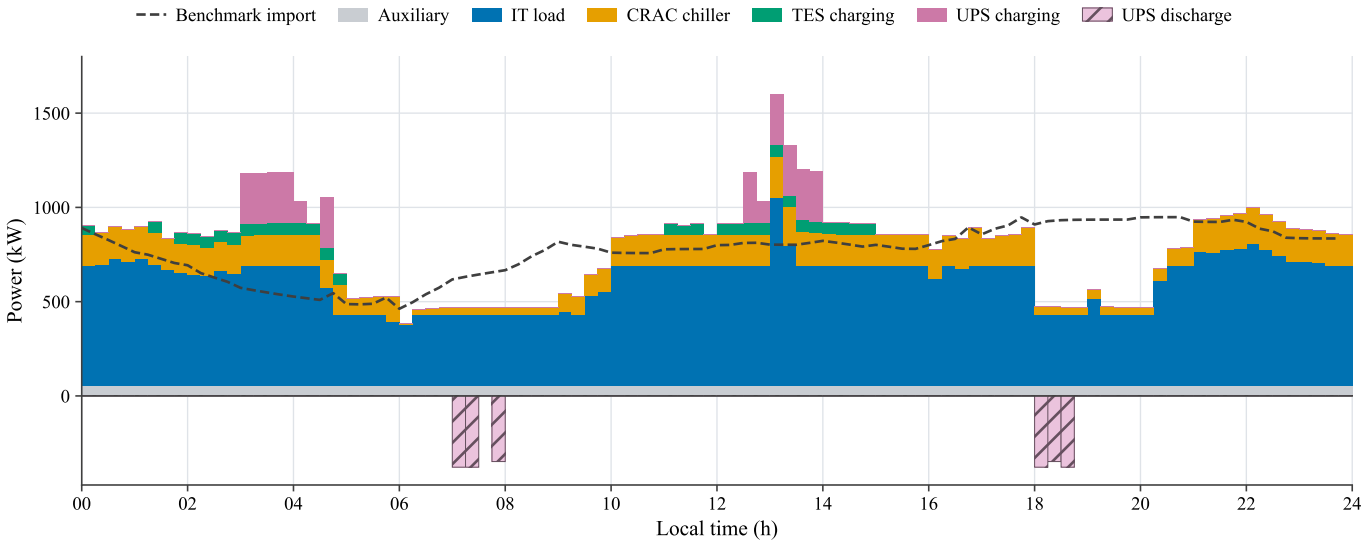


Fig. 6. Optimised component electrical power. Positive areas show consumption and charging; the hatched negative area denotes UPS discharge. Separate legend entries distinguish charging from discharging. TES discharge reduces chiller demand rather than supplying electricity. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

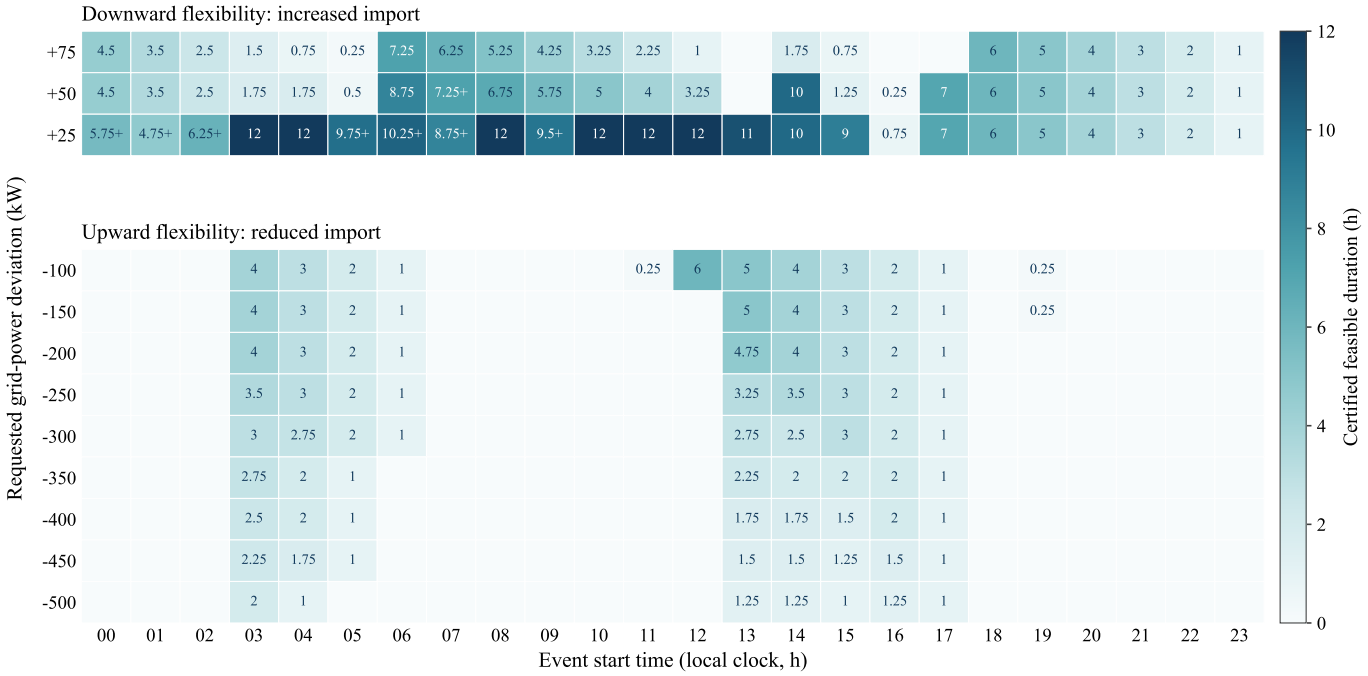


Fig. 7. Flexibility provision magnitude and duration. Cell values are certified feasible durations τ in hours for each start time t_0 (x-axis) and requested power deviation ΔP (y-axis). Positive ΔP (top three rows) denotes downward flexibility (load increase) and negative ΔP denotes upward flexibility (load reduction). A plus sign marks an unresolved next 15-minute trial. Blank cells have no positive duration certified by the search; twelve-hour and local-day-end values are assessment-capped. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

03:00, 04: 00, 08:00, 10:00, 11:00 and 12: 00. These changes reflect the specific baseline from which flexibility is provided: storage energy, unfinished work and thermal conditions differ between starts despite unchanged installed capacities. Many of the 100–200 kW reductions end at 07:00 or 18:00. At these times the undisturbed Scenario 2 schedule already uses UPS discharge to supply the entire IT load and has deferred all eligible work, so grid import falls to about 90 kW (Figure 6). A further reduction of 100 kW or more would require export, which the model does not permit; responses starting earlier therefore end at these times.

Another clear observation is that larger requested magnitudes generally have shorter certified durations, although plateaus occur. Larger deviations require more flexibility per time slot, while recovery also constrains the response. The positive and negative request ranges differ in magnitude, so their durations do not isolate direction from request size. Late-day durations are additionally limited by the end of the

selected day. Eight cells have an unresolved next trial: for example, a 25 kW increase at 06:00 is feasible for at least 10.25 h and a 50 kW increase at 07:00 for at least 7.25 h. A timeout does not establish an infeasible longer response, and even an adjacent infeasible duration does not prove global maximality under the moving recovery reference.

Each of the elements in Figure 7 can be decomposed and visualised as a stacked bar chart. A range of these stacked bar charts is shown in Figures 8 and 9. The plotted decomposition is an accepted feasible dispatch returned by the MILP for the given start time, flexibility magnitude and certified duration. The event-plus-recovery electricity cost objective selects among feasible dispatches, subject to the recorded solver gap; the component contributions shown should therefore be interpreted as one economically preferred realisation of the flexibility provision rather than a unique physical decomposition. In these plots, the duration is shown on the x-axis and the flexibility magnitude on the y-axis. Each stacked bar consists of the different sources of flexibility in the data centre, illustrating how much flexibility is being provided by each data centre asset throughout the flexibility provision. The dashed line shows the requested cumulative flexibility during the event. The target ends at the certified event duration. Grey regions contain no plotted recovery dispatch; recovery is checked through the terminal conditions. A plus sign beside a duration denotes an unresolved next trial. Notably, in a considerable number of time slots, certain data centre assets change their power consumption in the opposite direction to what is required. The other data centre assets compensate by adjusting their power consumption to a greater extent, ensuring that the desired net flexibility magnitude is achieved. This is visible in the charts, where the stacked bars include both negative and positive components. This effect clearly illustrates the data centre assets working in conjunction to achieve the desired net effect in the data centre.

Figure 8 presents the upward flexibility (negative ΔP) results as four stacked bar charts arranged into a 2-by-2 grid, in which the rows correspond to the requested flexibility magnitudes (-100 and -200 kW) and the columns to the start times (06:00 and 15:00). The colours show how much flexibility each data centre asset is providing. Figure 8 shows that τ depends strongly on ΔP and t_0. It is intuitive that the larger the flexibility magnitude, the shorter the duration it can be provided for. t_0 has a large impact on duration due to the specific baseline from which the flexibility is being provided. The left-hand side of Figure 8 shows the morning response: the selected 100 and 200 kW reductions at 06:00 are provided predominantly by additional UPS discharge for 1 h, with small reductions in IT and direct cooling demand. At 15:00, both reductions last 3 h and are provided mainly by IT workload deferral and reduced direct cooling. Additionally, cooling power reduction can take place when there is IT workload shifting as less heat is generated by the servers. TES is full and idle in the undisturbed trajectory at that afternoon start, so there is no TES charging to curtail. Both cases include some additional UPS discharge, averaging approximately 5 kW for the 200 kW request and 6 kW for the 100 kW request. Thus the participating resources change with the operating state; installed capacity alone does not determine which asset provides the response.

Figure 9 presents the downward flexibility (positive ΔP) results, with magnitudes of $+25$ and $+75$ kW and start times of 00:00 and 06:00. As with the upward flexibility charts, τ depends strongly on ΔP and t_0. The IT workload, CRAC, TES charging and UPS net exchange contribute through changes that may reinforce or oppose one another. Eligible work can be executed earlier than in Scenario 2, provided that it has already arrived. At 06:00, the 25 kW increase lasts at least 10.25 h and the 75 kW increase 7.25 h. One approach could be to include a 'charge up' time before the flexibility window where earlier time slots could reschedule their workload to later time slots in the flexibility window. This is akin to the recovery time slot implemented after the flexibility provision. Another notable result is the large positive and negative magnitudes exhibited by different data centre assets within the same time slot. As mentioned previously, these assets provide opposing flexibility responses that cancel each other out to achieve the desired effect. This allows one asset to provide flexibility while the other "recharges" its flexibility potential, thereby demonstrating the value of an integrated data centre model.

5.3 *Scenarios 1 and 2: Annual Electricity Costs*

The total annual base and optimised costs are GBP 550,503.59 and GBP 520,876.68, respectively, representing a GBP 29,626.91 cost saving. This demonstrates that, with an appropriate management algorithm, the modelled data centre flexibility assets can be used for electricity cost minimisation, yielding savings of 5.38%. The calculation includes conversion losses and preserves computational work, but excludes degradation, additional operating costs and service revenues.

Annual grid electricity increases from 6.718 to 6.789 GWh (approximately 1.05%), and peak import increases from 974.9 to 1,682.5 kW. Lower expenditure therefore does not imply lower electricity use or lower network peak demand. The valuation includes all observed 2025 prices, including 672 negative-price quarter-hours, whereas the selected physical-assessment day has none. It is a reference price electricity cost comparison rather than a complete site bill or market revenue calculation.

Table VI decomposes the total annual electricity cost of both scenarios by component. In the base case, IT load dominates the cost (74% of the total), followed by cooling (19%) and the constant auxiliary overhead (7%). Over 2025, the optimisation reduces the IT electricity cost term by GBP 16,722.17 and direct CRAC cooling by GBP 14,425.46. TES charging adds GBP 6,897.29, while net UPS exchange reduces expenditure by GBP 5,376.57; auxiliary expenditure is unchanged. These are signed accounting terms in the integrated schedule, not independent asset profits: TES benefits appear partly through reduced direct chiller demand, and the UPS term is charging minus discharging valued at interval prices. The net effect is the overall saving of GBP 29,626.91 (5.38%).

5.4 *Sensitivity Analysis*

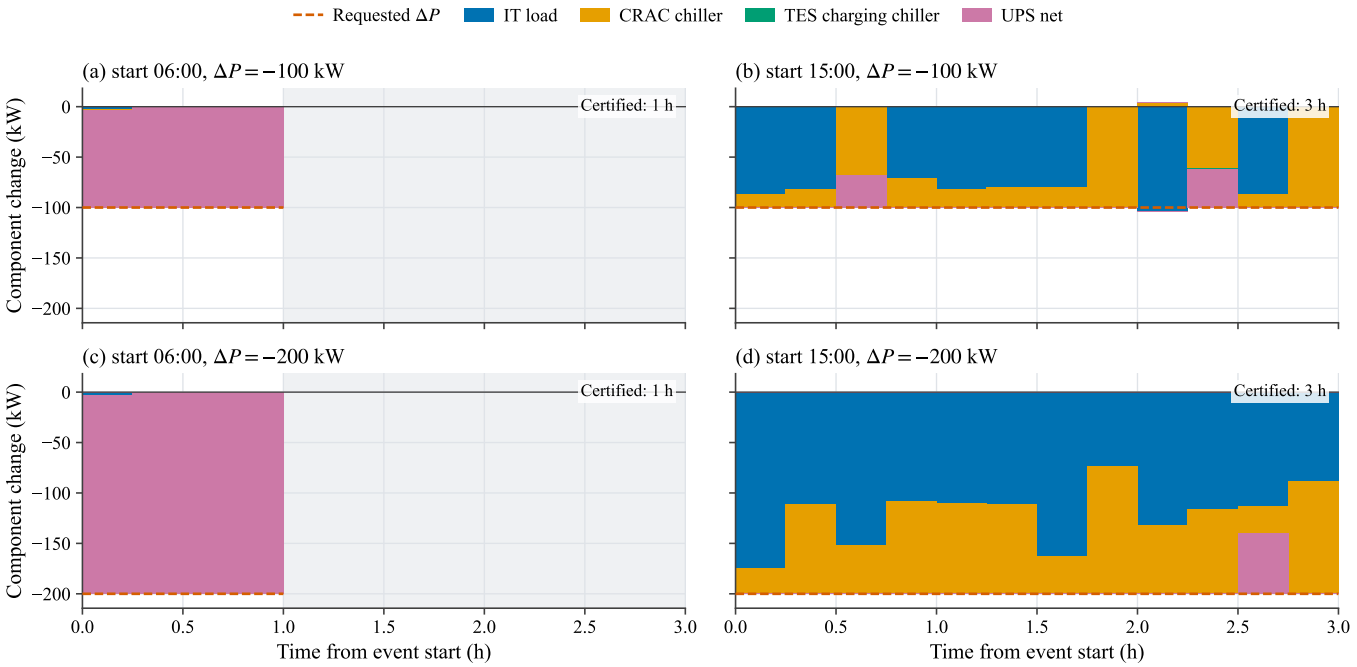


Fig. 8. Data centre component contributions to upward flexibility (negative ΔP). The four panels are arranged in a 2-by-2 grid: rows correspond to the requested flexibility magnitudes (-100 and -200 kW, top to bottom) and columns to the start times (06:00, left; 15:00, right). Panels (a)–(d) identify each start-time and magnitude combination. Colours show the flexibility contribution of each data centre asset. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

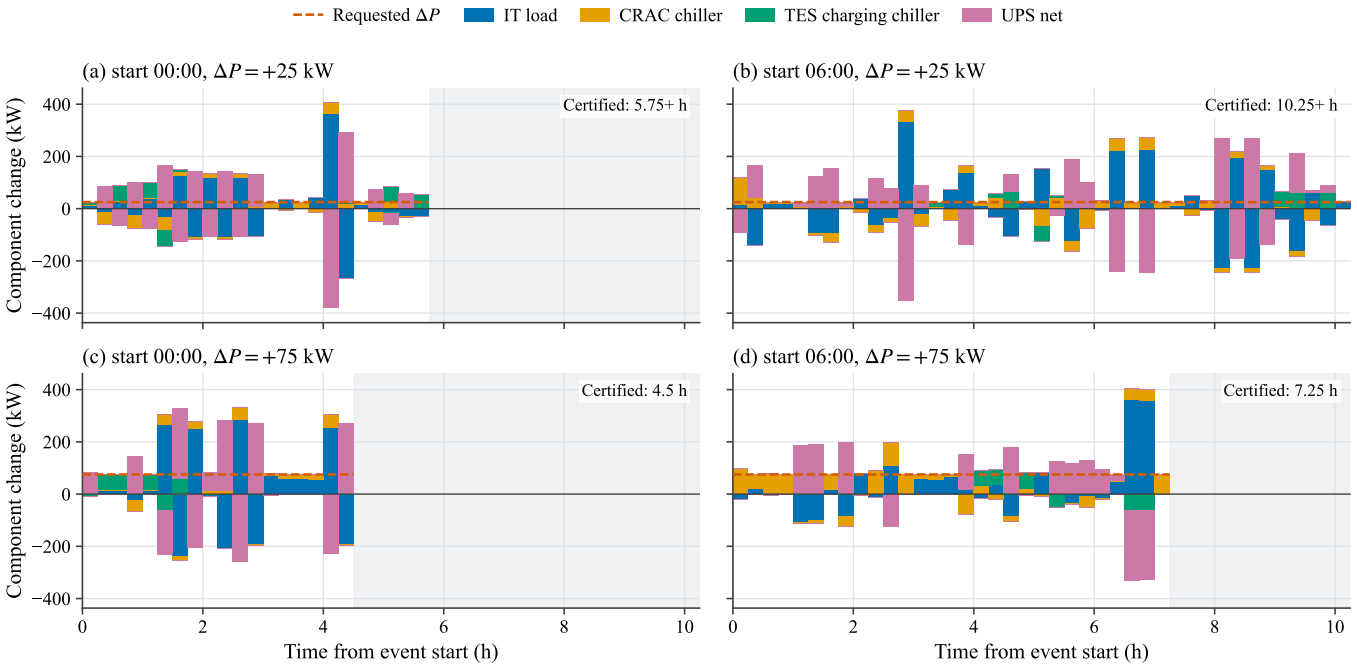


Fig. 9. Data centre component contributions to downward flexibility (positive ΔP). Rows correspond to the requested flexibility magnitudes ($+25$ and $+75$ kW, top to bottom) and columns to the start times (00:00, left; 06:00, right). Panels (a)–(d) identify each start-time and magnitude combination. Colours show the flexibility contribution of each data centre asset. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

Figure 10 summarises the OAT sensitivity analysis described in Section 4.4 using the full-year electricity cost saving. The base case yields a 5.38% saving; each curve shows the response when the corresponding resource parameter is varied between $0.5\times$ and $1.5\times$ its base value. All intermediate points are completed annual runs.

The cost saving is most sensitive to the flexible workload share among the resource variations tested: halving the share reduces the saving to 3.737%, while increasing it by $1.5\times$ raises the saving to 6.633%. At high multipliers the realised

Table VI
ANNUAL SIGNED ELECTRICITY COST COMPOSITION (GBP). CHANGES ARE COORDINATED MINUS REFERENCE; DISPLAYED VALUES ARE ROUNDED.

Component	Scenario 1	Scenario 2	Change
IT electrical demand	407,653	390,931	-16,722
CRAC chiller	105,337	90,911	-14,425
TES charging chiller	0	6,897	+6,897
UPS net grid effect	0	-5,377	-5,377
Auxiliary overhead	37,514	37,514	+0
Total settlement cost	550,504	520,877	-29,627

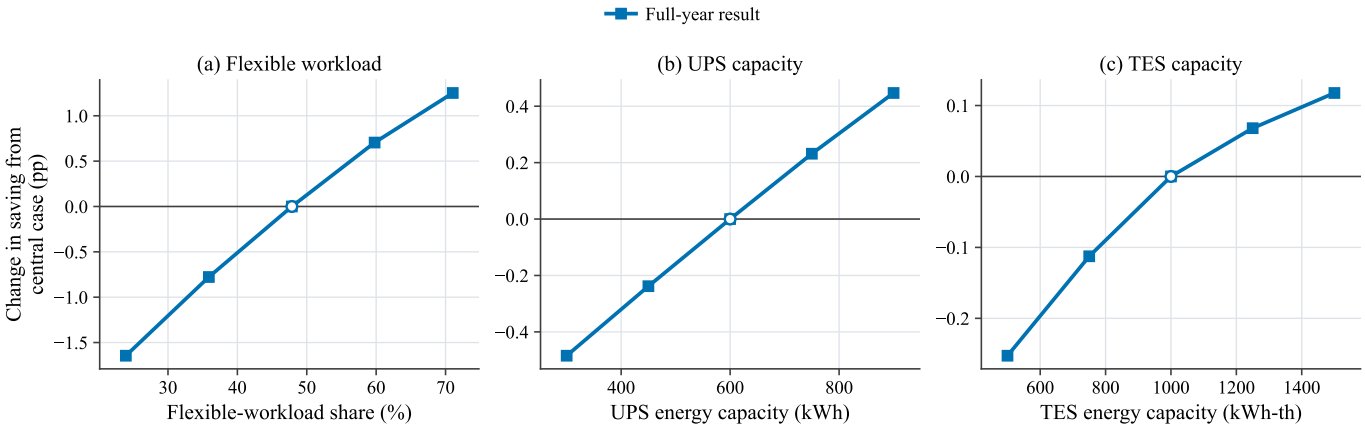


Fig. 10. Change in annual saving, in percentage points from the central case, for one-at-a-time resource variations. Panel (a) uses the realised flexible share after clipping. Note the different vertical scales. Plotted from the stored inputs or results using Python code prepared with assistance from ChatGPT Codex.

share saturates in slots where all arriving work is already flexible. Varying UPS energy capacity between $0.5\times$ and $1.5\times$ changes the saving between 4.897% and 5.829%, while varying TES capacity over the same range changes it between 5.129% and 5.499%. Storage power ratings remain fixed, so larger energy buffers do not remove charge/discharge or chiller limits. These results quantify dependence on the adopted workload and equipment assumptions rather than establishing robustness to all uncertainties or comparing resources at equal investment cost. The full annual results and solver exceptions are provided in Supplementary Sections S4–S5.

5.4.1 Incremental value of larger UPS batteries: The UPS sensitivity addresses whether additional energy capacity could be justified partly by its flexibility value. Increasing the central 600 kWh capacity to 750 and 900 kWh produces additional annual electricity cost savings of GBP 1,274.75 and GBP 2,459.89, respectively. With the reserve fraction fixed at 50%, the additions of 150 and 300 nameplate kWh increase the operational energy window by 75 and 150 kWh. Power ratings remain unchanged.

The calculation assumes that the additional electricity savings obtained for 2025 recur annually over the assumed battery lifetime. For a constant annual incremental saving ΔS, additional annual non-electricity cost O, real discount rate r and life n, the break-even incremental investment is

$$I_{\mathrm{BE}} = \frac{\Delta S - O}{\mathrm{CRF}(r,n)}, \qquad \mathrm{CRF}(r,n) = \frac{r(1+r)^n}{(1+r)^n - 1}. \quad (48)$$

The capital recovery factor annualises initial expenditure [39]. At an illustrative 8% real rate over ten years, with $O = 0$, the additional savings support investments of approximately GBP 8,554 and GBP 16,506, or GBP 57.02 and GBP 55.02 per added nameplate kWh. These are expenditure thresholds, not installed battery prices or an unconditional case for enlargement. Installation, degradation, O&M and replacement costs would need to be covered, and the retained single-year savings are not demonstrated lifetime cash flows: initial and terminal energy differ between capacities. Supplementary Section S6 gives the assumptions and alternative rate/life combinations. No revenue for the physical flexibility events is added to these electricity cost savings.

5.5 *Value of Integrated Co-optimisation*

The results allow the central question of what the integrated model reveals that separate, asset-level models would miss to be answered directly. First, the IT–cooling coupling: upward flexibility obtained by deferring IT workload produces a secondary reduction in cooling power because less heat is generated. A stand-alone IT flexibility model would omit the associated cooling power change, while a stand-alone cooling model with fixed IT demand would not represent the deferred heat load. Second, the change in resource participation visible in Figure 8: morning reductions rely on UPS discharge, whereas afternoon reductions rely mainly on IT deferral and cooling. This temporal coordination is only visible in a model that schedules both assets against a common grid power trajectory. Third, the TES buffering role: changes in direct CRAC cooling and TES charging interact with the other resources, allowing opposing responses in which one asset

"recharges" its flexibility potential while others over-deliver, as observed in Figures 8 and 9. These cross-asset interactions explain why the feasible duration envelope depends so strongly on the baseline operating state, and why a duration-aware, whole-facility formulation is required to characterise feasible grid-facing responses. These dispatch mechanisms do not quantify an integration premium over separately optimised subsystems, which would require an explicit comparator.

Finally, the scope of the grid interpretation should be noted. The model characterises the flexibility that the data centre can offer at its grid connection point; assessing the impact of the resulting load profile on upstream network stability or hosting capacity would require network topology, voltage and thermal limits, and the concurrent behaviour of other loads. The outputs of this work are therefore best interpreted as candidate flexibility offers to be evaluated within a system- or network-level study.

The workload split is context-dependent: applications, customer requirements, incentives and SLAs determine what may be shifted. The repeated daily workload, fixed ambient temperature and COP exclude seasonal and weekly facility changes. GPU power management, task dependencies and other forms of computational flexibility are outside the aggregate CPU-hour representation. The representative-day envelope does not establish annual availability, response speed below the quarter-hour resolution, or eligibility for a particular market product.

6. Conclusion

This paper presents an integrated, whole-facility framework that allows a data centre to act as a flexible power consumer with quantifiable flexibility provision. By co-optimising IT workload scheduling, UPS ESS dispatch, and thermodynamic cooling with thermal energy storage, a cost-minimising operating baseline was established through linked daily optimisation over 2025, achieving a 5.38 % reduction in annual electricity costs. A sensitivity analysis showed that the flexible workload share has the greatest effect among the tested resource variations, followed by UPS and TES energy capacity. Increasing UPS capacity from 600 to 900 kWh adds approximately GBP 2,460/year in savings, supporting a conditional break-even investment estimate. Building on this baseline, our core contribution is a duration-aware flexibility assessment that, for any start time (t_0) and requested power deviation (ΔP), identifies certified feasible durations (τ) of flexibility provision while enforcing operational constraints and the specified comparative recovery conditions. The resulting flexibility duration envelope reveals a strong temporal structure and a notable asymmetry. Upward flexibility is supplied predominantly by UPS discharge at some start times and by IT workload deferral with associated cooling reductions at others. In contrast, downward flexibility (load increase) combines changes in IT execution, cooling and storage, including earlier execution of work already available to process. This framework successfully quantifies power deviations and feasible durations under stated recovery rules, turning abstract potential into capabilities that can be assessed against market-specific requirements. Figure 7 provides a comprehensive set of results for the flexibility potential, detailing how the duration of flexibility provision changes depending on the magnitude required and specific start time. Results show a significant change in flexibility potential, with 100 kW of upward flexibility possible for 6 hours at noon and 1 hour at 06: 00. This change reflects the varying operating conditions of the baseline from which flexibility is being provided. The trade-off between magnitude and duration is equally clear: a 25 kW increase reaches the twelve-hour assessment cap at several start times, while larger tested requests generally have shorter certified durations. Cap-reaching results and unresolved adjacent trials are not intrinsic maximum-duration estimates. Interpreted per MW of IT capacity, these results indicate that the modelled data centre can sustain power deviations of the order of 10% of its IT capacity for multiple hours, with the feasible envelope determined by the start time and the requested magnitude.

Promising future work includes incorporating explicit revenue stacking, exploring a "pre-conditioning" window to assess additional IT participation in downward events, and aggregating facilities to a portfolio scale. Ultimately, the methodology presented offers a practical pathway for integrating data centres into power system operations, characterising candidate grid responses alongside annual electricity cost savings. Those savings do not imply lower energy consumption or peak import, and service revenues remain outside the present valuation.

CRediT authorship contribution statement

James Day: Writing – original draft, Investigation, Methodology, Conceptualization.
Mehmet Türker Takci: Writing – original draft, Investigation, Methodology, Conceptualization.
Meysam Qadrdan: Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All input data used in this study are publicly available or reported in full in this article. The electricity prices are the publicly available Low Carbon Contracts Company IMRP actuals [38]. The adopted workload profiles, deferral tranches and model parameters are given in Tables I, III and IV and Supplementary Section S1. The results supporting the findings are reported in the article and its supplementary information.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT Codex (OpenAI) and Claude Code (Anthropic)

to assist with manuscript preparation, editing and coding, including the Python code used to plot figures from the supplied numerical inputs and results, the workflow diagram and the redrawn facility schematic, as identified in the figure captions. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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REFERENCES

[1] Centrica, "LEM flexibility market platform design and trials report," Centrica, Tech. Rep., 2020, accessed 22 August 2025. [Online]. Available: https://www.centrica.com/media/4614/lem-flexibility-market-platform-design-and-trials-report.pdf

[2] European Environment Agency (EEA) and ACER, "Flexibility solutions to support a decarbonised and secure EU electricity system," EEA and ACER, Tech. Rep. EEA/ACER Report 09/2023, oct 2023, accessed 12 August 2025. [Online]. Available: https://www.eea.europa.eu/en/analysis/publications/flexibility-solutions-to-support

[3] International Energy Agency, "Energy and AI," International Energy Agency, Paris, Tech. Rep., 2025, accessed 14 October 2025. [Online]. Available: https://www.iea.org/reports/energy-and-ai

[4] National Energy System Operator (NESO), "Future Energy Scenarios 2025: Pathways to Net Zero," National Energy System Operator, Tech. Rep., 2025, accessed 23 July 2025. [Online]. Available: https://www.neso.energy/document/364541/download

[5] D. Koolen, M. D. Felice, and S. Busch, "Flexibility requirements and the role of storage in future European power systems," Publications Office of the European Union, Luxembourg, Tech. Rep. JRC130519, 2023. [Online]. Available: https://publications.jrc.ec.europa.eu/repository/handle/JRC130519

[6] Department for Energy Security and Net Zero, Ofgem and National Energy System Operator, "Clean flexibility roadmap," Department for Energy Security and Net Zero, Tech. Rep., jul 2025. [Online]. Available: https://www.gov.uk/government/publications/clean-flexibility-roadmap

[7] Y. Zhang, H. Tang, H. Li, and S. Wang, "Unlocking the flexibilities of data centers for smart grid services: optimal dispatch and design of energy storage systems under progressive loading," *Energy*, vol. 316, p. 134511, 2025. [Online]. Available: https://doi.org/10.1016/j.energy.2024.134511

[8] I. Alaperä, S. Honkapuro, and J. Paananen, "Data centers as a source of dynamic flexibility in smart grids," *Applied energy*, vol. 229, pp. 69–79, 2018. [Online]. Available: https://doi.org/10.1016/j.apenergy.2018.07.056

[9] H. Nosair and F. Bouffard, "Flexibility envelopes for power system operational planning," *IEEE Transactions on Sustainable Energy*, vol. 6, no. 3, pp. 800–809, jul 2015. [Online]. Available: https://doi.org/10.1109/TSTE.2015.2410760

[10] National Grid, "System needs and product strategy: UK electricity transmission," National Grid, Tech. Rep., jun 2017. [Online]. Available: https://www.nationalgrid.com/sites/default/files/documents/8589940795-System%20Needs%20and%20Product%20Strategy%20-%20Final.pdf

[11] L. Liu, X. Shen, Z. Chen, Q. Sun, and R. Wennersten, "Optimal energy management of data center micro-grid considering computing workloads shift," *IEEE Access*, vol. 12, pp. 102 061–102 075, 2024. [Online]. Available: https://doi.org/10.1109/ACCESS.2024.3432120

[12] Y. Cao, M. Cheng, S. Zhang, H. Mao, P. Wang, C. Li, Y. Feng, and Z. Ding, "Data-driven flexibility assessment for internet data center towards periodic batch workloads," *Applied Energy*, vol. 324, p. 119665, oct 2022. [Online]. Available: https://doi.org/10.1016/j.apenergy.2022.119665

[13] M. S. Misaghian, G. Tardioli, A. G. Cabrera, I. Salerno, D. Flynn, and R. Kerrigan, "Assessment of carbon-aware flexibility measures from data centres using machine learning," *IEEE Transactions on Industry Applications*, vol. 59, no. 1, pp. 70–80, jan 2023. [Online]. Available: https://doi.org/10.1109/TIA.2022.3213637

[14] A. Radovanović, R. Koningstein, I. Schneider, B. Chen, A. Duarte, B. Roy, D. Xiao, M. Haridasan, P. Hung, N. Care *et al.*, "Carbon-aware computing for datacenters," *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1270–1280, 2023. [Online]. Available: https://doi.org/10.1109/TPWRS.2022.3173250

[15] I. Alaperä, J. Paananen, K. Dalen, and S. Honkapuro, "Fast frequency response from a UPS system of a data center, background, and pilot results," in *2019 16th International Conference on the European Energy Market (EEM)*. IEEE, 2019, pp. 1–5. [Online]. Available: https://doi.org/10.1109/EEM.2019.8916344

[16] J. Roach, "Microsoft datacenter batteries to support growth of renewables on the power grid," Microsoft Source, 2022, accessed 11 May 2025. [Online]. Available: https://news.microsoft.com/source/features/sustainability/microsoft-datacenter-batteries-to-support-growth-of-renewables-on-the-power-grid/

[17] V.-G. Anghel, "What you need to know about grid-interactive data centers," DatacenterDynamics, 2023, accessed 21 May 2025. [Online]. Available: https://www.datacenterdynamics.com/en/analysis/what-you-need-to-know-about-grid-interactive-data-centers/

[18] M. Zhao and X. Wang, "A synthetic approach for datacenter power consumption regulation towards specific targets in smart grid environment," *Energies*, vol. 14, no. 9, p. 2602, 2021. [Online]. Available: https://doi.org/10.3390/en14092602

[19] ASHRAE TC 9.9, "Data center power equipment thermal guidelines and best practices," ASHRAE, Tech. Rep., 2016.

[20] M. T. Takci, M. Qadrdan, J. Summers, and J. Gustafsson, "Data centres as a source of flexibility for power systems," *Energy Reports*, vol. 13, pp. 3661–3671, 2025.

[21] R. Basmadjian, "Flexibility-based energy and demand management in data centers: a case study for cloud computing," *Energies*, vol. 12, no. 17, p. 3301, 2019. [Online]. Available: https://doi.org/10.3390/en12173301

[22] H. Du, X. Zhou, N. Nord, Y. Carden, P. Cui, and Z. Ma, "A new framework for evaluating and enhancing the performance of district heating systems integrated with data centres using short-term thermal energy storage," *Energy*, p. 134934, 2025. [Online]. Available: https://doi.org/10.1016/j.energy.2025.134934

[23] Y. Fu, X. Han, K. Baker, and W. Zuo, "Assessments of data centers for provision of frequency regulation," *Applied Energy*, vol. 277, p. 115621, nov 2020. [Online]. Available: https://doi.org/10.1016/j.apenergy.2020.115621

[24] S. Xiang, Y. Xiang, Y. Lu, Y. Guo, Z. Tan, and Y. Wang, "Modeling and optimization of data center energy consumption," in *2023 Panda Forum on Power and Energy (PandaFPE)*. IEEE, 2023, pp. 544–549. [Online]. Available: https://doi.org/10.1109/PandaFPE58872.2023.10249234

[25] H. Xu, Y. Li, H. Zhu, J. Wang, and C. Hou, "Data center demand response potential assessment considering multiple types of flexible resources," in *Proceedings of SPIE – Fourth International Conference on Computer Science and Communication Technology (ICCSCT 2023)*, vol. 12918, oct 2023. [Online]. Available: https://doi.org/10.1117/12.3009383

[26] T. Cioara, I. Anghel, M. Bertoncini, I. Salomie, D. Arnone, M. Mammina, T.-H. Velivassaki, and M. Antal, "Optimized flexibility management enacting data centres participation in smart demand response programs," *Future Generation Computer Systems*, vol. 78, pp. 330–342, 2018. [Online]. Available: https://doi.org/10.1016/j.future.2016.05.010

[27] L. A. Barroso, U. Hölzle, and P. Ranganathan, *The Datacenter as a Computer: Designing Warehouse-Scale Machines*, 3rd ed., ser. Synthesis Lectures on Computer Architecture. Morgan & Claypool Publishers, 2018.

[28] M. Dayarathna, Y. Wen, and R. Fan, "Data center energy consumption modeling: A survey," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 732–794, 2016. [Online]. Available: https://doi.org/10.1109/COMST.2015.2481183

[29] C. Zaloumis, "Are your data centers keeping you from sustainability?" IBM Think, accessed 12 July 2025. [Online]. Available: https://www.ibm.com/think/insights/are-your-data-centers-keeping-you-from-sustainability

[30] A. Ghia, "Capturing value through IT consolidation and shared services," *McKinsey on Government*, pp. 18–23, 2011, autumn issue. [Online]. Available: https://www.mckinsey.com/~/media/mckinsey/

dotcom/client_service/Public%20Sector/PDFS/McK%20on%20Govt/IT%20Challenge%20and%20opportunity/MOG7_Consolidation.ashx

[31] M. Rasheduzzaman, M. A. Islam, T. Islam, T. Hossain, and R. M. Rahman, "Task shape classification and workload characterization of google cluster trace," in *2014 IEEE International Advance Computing Conference (IACC)*, 2014, pp. 893–898.

[32] S. Zhou, M. Zhou, Z. Wu, Y. Wang, and G. Li, "Energy-aware coordinated operation strategy of geographically distributed data centers," *International Journal of Electrical Power & Energy Systems*, vol. 159, p. 110032, aug 2024. [Online]. Available: https://doi.org/10.1016/j.ijepes.2024.110032

[33] Z. Liu, Y. Chen, C. Bash, A. Wierman, D. Gmach, Z. Wang, M. Marwah, and C. Hyser, "Renewable and cooling aware workload management for sustainable data centers," *ACM SIGMETRICS Performance Evaluation Review*, vol. 40, no. 1, pp. 175–186, jun 2012. [Online]. Available: https://doi.org/10.1145/2318857.2254779

[34] J. v. Kistowski, H. Block, J. Beckett, K.-D. Lange, J. A. Arnold, and S. Kounev, "Analysis of the influences on server power consumption and energy efficiency for CPU-intensive workloads," in *Proceedings of the 6th ACM/SPEC International Conference on Performance Engineering*, 2015, pp. 223–234. [Online]. Available: https://doi.org/10.1145/2668930.2688057

[35] M. T. Takcı, T. Gözel, and M. H. Hocaoglu, "Modelling, analysis, and improvement of energy consumption in data centres via demand side management," in *Energy Efficiency of Modern Power and Energy Systems*, S. H. E. A. Aleem, M. E. Balci, and M. J. H. Rawa, Eds. Elsevier, 2024, pp. 73–99. [Online]. Available: https://doi.org/10.1016/B978-0-443-21644-2.00005-1

[36] M. U. Hashmi, D. V. Hertem, A. van der Meer, and A. Keane, "Linear energy storage and flexibility model with ramp rate, ramping, deadline and capacity constraints," sep 2024. [Online]. Available: https://arxiv.org/abs/2409.08084

[37] L. Cupelli, T. Schütz, P. Jahangiri, M. Fuchs, A. Monti, and D. Müller, "Data center control strategy for participation in demand response programs," *IEEE Transactions on Industrial Informatics*, vol. 14, no. 11, pp. 5087–5099, 2018. [Online]. Available: https://doi.org/10.1109/TII.2018.2806889

[38] Low Carbon Contracts Company, "IMRP actuals," LCCC Data Portal, 2026. [Online]. Available: https://dp.lowcarboncontracts.uk/dataset/imrp-actuals

[39] National Renewable Energy Laboratory, "System Advisor Model: LCOE Calculator, capital recovery factor," SAM Help, n.d., accessed 8 September 2026. [Online]. Available: https://samrepo.nlr.gov/help/fin_lcoefcr.html